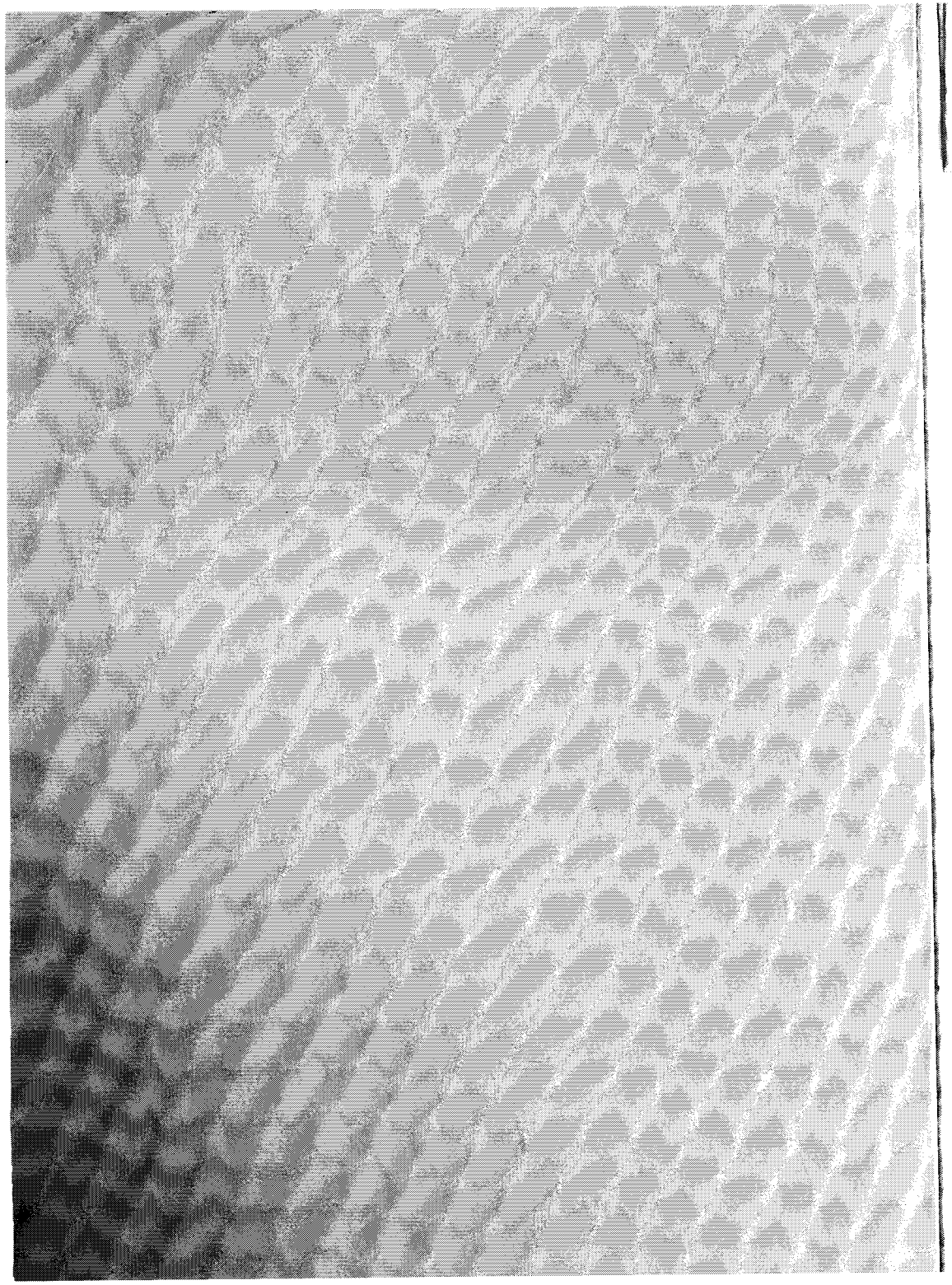


NOTES ON
COBORDISM THEORY

BY
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Preface

These notes represent the outgrowth of an offer by Princeton University to let me teach a graduate level course in cobordism theory. Despite the fact that cobordism notions appear in the earliest literature of algebraic topology, it has only been since the work of Thom in 1954 that more than isolated results have been available. Since that time the growth of this area has been phenomonal, but has largely taken the form of individual research papers. To a certain extent, the nature of cobordism as a classificational tool has led to the study of many individual applications rather than the development of a central theory. In particular, there is no complete exposition of the fundamental results of cobordism theory, and it is hoped that these notes may help to fill this gap.

Being intended for graduate and research level work, no attempt is made here to use only elementary ideas. In particular, it is assumed that the reader knows algebraic topology fairly thoroughly, with cobordism being treated here as an application of topology. In many cases this is not the fashion in which development took place, for ideas from cobordism have frequently led to new methods in topology itself.

An attempt has been made to provide references to the sources of most of the ideas used. Although the main ideas of these sources are followed closely, the details have frequently been modified considerably. Thus the reader may find it helpful to refer to the original papers to find other methods which are useful. For example, the Adams spectral sequence gives a powerful computational tool which has been used in determining some theories and which facilitates low dimensional calculations, but is never used here. Many of the ideas which appear are of the "well known to workers in the field - but totally unavailable" type and a few ideas are my own.

The pattern of exposition follows my own prejudices, and may be roughly described as follows. There are three central ideas in cobordism theory:

1) Definition of the cobordism groups,

2) Reduction to a homotopy problem, and

3) Establishing cobordism invariants.

This material is covered in the first three chapters. Beyond that point, one must become involved with the peculiarities of the individual cobordism problem. This is begun in the fourth chapter with a survey of the literature, followed by detailed discussion of specific cobordism theories in the later chapters. Finally, two appendices covering advanced calculus and differential topology are added, this material being central to the 'reduction to a homotopy problem' but being of such a nature as to overly delay any attempt to get rapidly to the ideas of cobordism.

I am indebted to many people for leading me to this work and developing my ideas in this direction. Especially, I am indebted to Greg Brumfiel, Peter Landweber, and Larry Smith for numerous suggestions in preparing these notes, and to Mrs. Barbara Duld for typing. I am indebted to Princeton University and the National Science Foundation for financial support. Finally, I am indebted to my wife for putting up with the foul moods I developed during this work.

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CHAPTER I

Introduction - Cobordism Categories

In order to place the general notion of cobordism theory in mathematical perspective recall that differential topology is the study of the category of differentiable manifolds and differentiable maps, primarily in relation to the category of topological spaces and continuous maps. From a slightly less theoretical point of view, it is the study of differentiable manifolds by topologists using any methods they can find. The guiding principle is that one does not study imposed structures such as Riemannian metrics or connections and this distinguishes differential topology from differential geometry.

As in any subject, the primary problem is classification of the objects within isomorphism and determination of effective and computable invariants to distinguish the isomorphism classes. In the case of differentiable manifolds this problem is not solvable, since for any finitely presented group S one can construct a four dimensional manifold $M(S)$ with fundamental group S in such a way that $M(S)$ and $M(T)$ will be homeomorphic if and only if S and T are isomorphic, but one cannot solve the word problem to determine whether two finitely presented groups are isomorphic (Markov [76]). In special cases one can solve the problem, but cobordism theory works in another way - by introducing an equivalence relation much weaker than isomorphism.

Briefly, two manifolds without boundary are called 'cobordant' if their disjoint union is the boundary of some manifold. It is worthwhile to note that every manifold M with empty boundary is the boundary of $M \times [0,\infty)$. To get a nontrivial theory it is standard to restrict attention to compact manifolds.

The first description of this equivalence relation was by H. Poincaré: Analysis Situs, Journal de l'École Polytechnique, 1 (1895), 1-121 (section 5, Homologies). His concept of homology is basically the same as the concept of cobordism used today.

The next development of cobordism theory was by L. S. Pontrjagin: Characteristic cycles on differentiable manifolds, Math. Sbor. (N.S.), 21 (63) (1947), 233-284 (Amer. Math. Soc. translations, series 1, no. 32). This paper shows that the characteristic numbers of a closed manifold vanish if the manifold is a boundary (providing the invariants for classification).

The cobordism classification of manifolds is reasonably elementary in dimensions 0,1, and 2, since manifolds are themselves classified in these dimensions. Using geometric methods the cobordism classification problem in dimension 3 was solved by V. A. Rohlin: A 3-dimensional manifold is the boundary of a 4-dimensional manifold, Doklady Akad. Nauk. S.S.S.R., 81 (1951), 355.

The first application of cobordism was by L. S. Pontrjagin: Smooth manifolds and their applications in homotopy theory, Trudy Mat. Inst. im Steklov no. 45, Izdat. Akad. Nauk. S.S.S.R. Moscow, 1955 (Amer. Math. Soc. translations, series 2, vol. 11, 1959). Pontrjagin attempted to study the stable homotopy groups of spheres as the groups of cobordism classes of 'framed' manifolds. This amounts to the equivalence of a homotopy problem and a cobordism problem. The lack of knowledge of manifolds has prevented this from being of use in solving the homotopy problem.

The major development of cobordism theory is the paper of R. Thom: Quelques propriétés globales des variétés differentiables, Comm. Math. Helv., 28 (1954), 17-86. This paper showed that the problem of cobordism is

equivalent to a homotopy problem. For many of the interesting manifold classification questions the resulting homotopy problem turns out to be solvable. Thus, Thom brought the Pontrjagin technique to the study of manifolds, largely reversing the original idea.

For a brief sketch of cobordism theory there are three survey articles of considerable interest. For an insight into the early development of the theory (up through Thom's work) see V. A. Rohlin: Intrinsic homology theories, Uspekhi Mat. Nauk., 14 (1959), 3-20 (Amer. Math. Soc. translations, series 2, 30 (1963), 255-271). A short article which covers many of the examples of cobordism classification problems is J. Milnor: A survey of cobordism theory, Enseignement Mathematique, 8 (1962), 16-23. Contained in the survey of differential topology by C. T. C. Wall: Topology of smooth manifolds, Journal London Math. Soc., 40 (1965), 1-20, is a discussion of representative cobordism theories, with outlines of the methods by which these problems are solved.

Cobordism Categories

In order to formalize the notion of cobordism theory, it seems useful to set up a 'general nonsense' situation. As motivation, one may consider the properties of differentiable manifolds.

Let $\mathcal{D}$ denote the category whose objects are compact differentiable manifolds with boundary (of class C^∞) and whose maps are the differentiable maps (again C^∞) which take boundary into boundary. This category has finite sums given by the disjoint union and has an initial object given by the empty manifold. For each object of $\mathcal{D}$ one has its boundary, again an object of $\mathcal{D}$, and for each map the restriction of it to the boundary.

Further, the boundary of the boundary is always empty. This defines an additive functor $\partial : \mathcal{P} \longrightarrow \mathcal{P}$. For any manifold M, the boundary of M is a subset whose inclusion is a differentiable map $i(M) : \partial M \longrightarrow M$. This inclusion gives a natural transformation $i : \partial \longrightarrow I$ of additive functors, $I : \mathcal{P} \longrightarrow \mathcal{P}$ being the identity functor. Finally, the Whitney imbedding theorem shows that each differentiable manifold is isomorphic to a submanifold of countable dimensional Euclidean space. Thus $\mathcal{P}$ has a small subcategory $\mathcal{P}_0$ (submanifolds of R^∞) such that each object of $\mathcal{P}$ is isomorphic to an object of $\mathcal{P}_0$.

Abstracting these properties, one has:

Definition: A cobordism category $(\mathcal{C},\partial,i)$ is a triple in which:

1) $\mathcal{C}$ is a category having finite sums and an initial object;
2) $\partial : \mathcal{C} \longrightarrow \mathcal{C}$ is an additive functor such that for each object X of $\mathcal{C}$, $\partial\partial(X)$ is an initial object;
3) $i : \partial \longrightarrow I$ is a natural transformation of additive functors from ∂ to the identity functor I; and
4) There is a small subcategory $\mathcal{C}_0$ of $\mathcal{C}$ such that each object of $\mathcal{C}$ is isomorphic to an object of $\mathcal{C}_0$.

As noted in motivating this definition, $(\mathcal{P},\partial,i)$ is a cobordism category. There are many more examples, and in fact the purpose of cobordism theory is to study the interesting examples. The precise choice of this formulation is based, somewhat vaguely, on the definition of 'adjoint functors'.

The purpose of this definition is not to establish a general nonsense structure; rather the definition will be used to follow the framework of previously developed theory and to try to unify the ideas. To begin, one has in any cobordism category the idea of a 'cobordism relation'.

Definition: If $(\mathcal{C},\partial,i)$ is a cobordism category, one says that the objects X and Y of $\mathcal{C}$ are cobordant if there exist objects U and V of $\mathcal{C}$ such that the sum of X and ∂U is isomorphic to the sum of Y and ∂V. This will be written $X \equiv Y$.

One has easily:

Proposition:

a) $\equiv$ is an equivalence relation on the objects of $\mathcal{C}$.

b) $X \equiv Y$ implies $\partial X \tilde{=} \partial Y$.

c) For all X, $\partial X \equiv \emptyset$ where $\emptyset$ is an initial object.

d) If $X \equiv X'$, $Y \equiv Y'$ and Z and Z' are sums of the pairs (X,Y) and (X',Y') respectively, then $Z \equiv Z'$.

Proof:

a) $X + \partial\emptyset \tilde{=} X + \partial\emptyset$;

$X + \partial U \tilde{=} Y + \partial V \Longrightarrow Y + \partial V = X + \partial U$; and

$X + \partial U \tilde{=} Y + \partial V$, $Y + \partial W \tilde{=} Z + \partial T$ implies $X + \partial(U+W) \tilde{=} X + \partial U + \partial W \tilde{=} Y + \partial V + \partial W \tilde{=} Z + \partial V + \partial T \tilde{=} Z + \partial(V+T)$.

b) $X + \partial U \tilde{=} Y + \partial V$ implies $\partial X \tilde{=} \partial X + \emptyset \tilde{=} \partial X + \partial\partial U \tilde{=} \partial Y + \partial\partial V \tilde{=} \partial Y + \emptyset \tilde{=} \partial Y$.

c) $\partial X + \partial\emptyset \tilde{=} \emptyset + \partial X$ since $\partial\emptyset$ is initial.

d) $X + \partial U \tilde{=} X' + \partial U'$, $Y + \partial V \tilde{=} Y' + \partial V'$ gives $Z + \partial(U+V) \tilde{=} Z' + \partial(U'+V')$

NOTE: In all of the above $A + B$ denotes an object which is a sum for A and B. **

Note: If one is unhappy with equivalence relations on a category, one may reduce to considering $\equiv$ as an equivalence relation on the set of isomorphism classes of objects of $\mathcal{C}$. This is the reason for the assumption about existence of $\mathcal{C}_0$.

Definition: An object X of $\mathcal{C}$ is closed if ∂X is an initial object. An object X of $\mathcal{C}$ bounds if $X \equiv \emptyset$ where $\emptyset$ is an initial object.

Proposition:

a) X closed and $Y \equiv X$ implies Y closed.

b) X and X' closed implies their sum is closed.

c) X bounds implies X is closed.

d) X and Y bound implies their sum bounds.

e) X bounds and $Y \equiv X$ implies Y bounds.

Proof:

a) follows directly from b) above.

b) $\partial X \tilde{=} \emptyset$, $\partial X' \tilde{=} \emptyset$ implies $\partial(X+X') \tilde{=} \emptyset + \emptyset \tilde{=} \emptyset$.

c) $X \equiv \emptyset$ implies $\partial X \tilde{=} \partial\emptyset \tilde{=} \emptyset$.

d) $X \equiv \emptyset$, $Y \equiv \emptyset$ implies $X + Y \equiv \emptyset + \emptyset \tilde{=} \emptyset$.

e) is immediate since $\equiv$ is an equivalence relation. **

Proposition: The set of equivalence classes of closed objects of $\mathcal{C}$ (under $\equiv$) has an operation induced by the sum in $\mathcal{C}$. This operation is associative, commutative, and has a unit (the class of any object which bounds).

Proof: The existence of $\mathcal{C}_0$ implies that the equivalence classes form a set. That the sum in $\mathcal{C}$ gives rise to an operation on this set follows immediately from the propositions. Associativity and commutativity hold for isomorphism classes of objects, hence also here. **

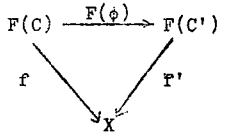
Definition: The cobordism semigroup of the cobordism category $(\mathcal{C},\partial,i)$ is the set of equivalence classes of closed objects of $\mathcal{C}$ with the operation induced by the sum in $\mathcal{C}$. This semigroup will be denoted $\Omega(\mathcal{C},\partial,i)$.

Remarks: 1) $\Omega(\mathcal{C},\partial,i)$ may also be described as the semigroup of isomorphism classes of closed objects of $\mathcal{C}$ modulo the sub-semigroup of isomorphism classes of objects which bound.

2) The semigroup $\Omega(\mathcal{O},\partial,i)$ is quite easily identifiable as Thom's cobordism group $\mathcal{N}_*$ of unoriented cobordism classes of closed manifolds. In order to clarify this slightly, in the usual expression for equivalence one has X equivalent to Y if there is a V with $\partial V = X \cup Y$. Then $X \cup \partial V \tilde{=} Y \cup \partial(X\times I)$ giving $X \equiv Y$. The implication $X \cup \partial U \tilde{=} Y \cup \partial V$ implies $X \cup Y = \partial T$ is an easy geometric argument by looking at components and piecing together manifolds with boundary by means of tubular neighborhoods of their boundary components.

Within the literature of cobordism there are a few standard constructions performed. These may be generalized to the categorical situation as will now be shown.

Construction I: Let $(\mathcal{C},\partial,i)$ be a cobordism category, $\mathcal{X}$ a category with finite sums and an initial object, and $F : \mathcal{C} \longrightarrow \mathcal{X}$ an additive functor. For any object X of $\mathcal{X}$, form a category $\mathcal{C}/X$ whose objects are pairs (C,f) with C an object of $\mathcal{C}$ and $f \in \mathrm{Map}(F(C),X)$ and whose maps are given by letting $\mathrm{Map}((C,f),(C',f'))$ be the set of maps $\phi \in \mathrm{Map}(C,C')$ such that the diagram



commutes.

If $\emptyset$ is an initial object of $\mathcal{C}$ and $\phi : F(\emptyset) \longrightarrow X$ is the unique map, then $(\emptyset,\phi)$ is an initial object of $\mathcal{C}/X$. If (D,g) and (D',g') are objects of $\mathcal{C}/X$ and $D+D'$ is a sum for D and D' in $\mathcal{C}$, then $F(D+D')$ is a sum for $F(D)$ and $F(D')$ in $\mathcal{X}$. The maps g and g' give a well defined map $g+g' : F(D+D') \longrightarrow X$, and $(D+D',g+g')$ is the sum of (D,g) and (D',g') in $\mathcal{C}/X$.

Let $\tilde{\partial}(C,f) = (\partial C, f\circ F(i_C))$ and $\tilde{\partial}(\phi) = \phi\circ i_C$ to define the functor $\tilde{\partial} : \mathcal{C}/X \longrightarrow \mathcal{C}/X$. Define the natural transformation $\tilde{i} : \tilde{\partial} \longrightarrow I$ by $\tilde{i}_{(C,f)} = i_C : \partial C \longrightarrow C$.

Then $(\mathcal{C}/X,\tilde{\partial},\tilde{i})$ is a cobordism category.

Remarks: 1) This is the algebraic-geometric (Grothendieck style) notion of the category of objects over a given object.

2) If one begins with the category $(\mathcal{C},\partial,i)$ and takes $F : \mathcal{C} \longrightarrow \mathcal{X}$ to be the forgetful functor to the category of topological spaces and continuous maps, then $\Omega(\mathcal{C}/X,\tilde{\partial},\tilde{i})$ is the unoriented bordism group $\mathcal{T}_*(X)$ as originally formulated by M. F. Atiyah: Bordism and cobordism, Proc. Camb. Phil. Soc., 57 (1961), 200-208.

Construction II: Let $\mathcal{A}$ be a small category, $(\mathcal{C},\partial,i)$ a cobordism category, and let $\mathrm{Fun}(\mathcal{A}, \mathcal{C})$ be the category whose objects are functors $\phi : \mathcal{A} \longrightarrow \mathcal{C}$ and whose maps are the natural transformations.

If $\emptyset$ is an initial object of $\mathcal{C}$, the constant functor $\Phi : \mathcal{A} \longrightarrow \mathcal{C} : A \longrightarrow \emptyset$ is an initial object of $\mathrm{Fun}(\mathcal{A},\mathcal{C})$. If $F,G : \mathcal{A} \longrightarrow \mathcal{C}$ are functors, let $H : \mathcal{A} \longrightarrow \mathcal{C}$ by letting $H(A)$ be a sum for $F(A)$ and $G(A)$ and let $(j_F)_A = j_{F(A)} : F(A) \longrightarrow H(A)$ and $(j_G)_A = j_{G(A)} : G(A) \longrightarrow H(A)$ be the maps exhibiting $H(A)$ as the sum. Then j_F and j_G are natural transformations which exhibit H as a sum for F and G.

Let $\tilde{\partial} : \mathrm{Fun}(\mathcal{A},\mathcal{C}) \longrightarrow \mathrm{Fun}(\mathcal{A},\mathcal{C}) : F \longrightarrow \partial\circ F : \lambda \longrightarrow \partial(\lambda)$ and let $\tilde{i} : \tilde{\partial} \longrightarrow I$ be the natural transformation given by the map $\tilde{i}_F : \partial\circ F \longrightarrow F$ whose evaluation at any object A of $\mathcal{A}$ is $i_{F(A)} : \partial(F(A)) \longrightarrow F(A)$.

Then $(\mathrm{Fun}(\mathcal{A},\mathcal{C}),\tilde{\partial},\tilde{i})$ is a cobordism category.

Remark: Many standard examples fit this construction. Suppose $\mathcal{A}$ is the category with one object A whose maps are a finite group $G = \mathrm{Map}(A,A)$. A functor $F : \mathcal{A} \longrightarrow \mathcal{D}$ is given by selecting a manifold $X = F(A)$ and a homomorphism $G \longrightarrow \mathrm{Map}(X,X)$. Since G is finite, the induced map $G \times X \longrightarrow X$ is a differentiable action of G on X. Thus $\Omega(\mathrm{Fun}(\mathcal{A},\mathcal{D}),\tilde{\partial},\tilde{i})$ is the unoriented cobordism group of (unrestricted) G-actions as defined by P. E. Conner and E. E. Floyd: "Differentiable Periodic Maps", Springer, Berlin, 1964 (section 21).

Relative Cobordism

In order to study the relationship between two cobordism categories it is convenient to have available a 'relative cobordism' semigroup. In the geometric case this is made possible by joining together two manifolds with the same boundary to form a closed manifold. In the categorical situation, the idea is to replace a pair of objects having the same boundary

by a pair of closed objects. For this one needs the idea of the Grothendieck group construction.

Recall that for any category with finite sums for which the isomorphism classes of objects form a set, $\mathcal{X}$, one defines $K(\mathcal{X})$, the Grothendieck group of $\mathcal{X}$, to be the set of equivalence classes of pairs (X,X') of objects of $\mathcal{X}$, where (X,X') is equivalent to (Y,Y') if there is an object A of $\mathcal{X}$ such that $X + Y' + A \cong X' + Y + A$. $K(\mathcal{X})$ is an abelian group under the operation induced by the sum in $\mathcal{X}$.

Let $(\mathcal{C},\partial,i)$ and $(\mathcal{C}',\partial',i')$ be two cobordism categories, $F : \mathcal{C} \longrightarrow \mathcal{C}'$ an additive functor, and $t : \partial'F \cong F\partial$ a natural equivalence of additive functors such that the diagram

$$\begin{array}{ccc} \partial'F(A) & \xrightarrow{t(A)} & F(\partial A) \\ & {\scriptstyle i'_{F(A)}} \searrow \quad \swarrow {\scriptstyle F(i_A)} & \\ & F(A) & \end{array}$$

commutes. Let $\mathcal{P}$ be the category whose objects are triples (X,Y,f) with $X \in \mathcal{C}'$, $Y \in \mathcal{C}$, Y closed, and $f : \partial'X \longrightarrow FY$ an isomorphism and with $\mathrm{Map}((X,Y,f),(X',Y',f'))$ the set of $(\phi,\psi) \in \mathrm{Map}(X,X') \times \mathrm{Map}(Y,Y')$ such that

$$\begin{array}{ccc} \partial'X & \xrightarrow{f} & FY \\ {\scriptstyle \partial'\phi}\downarrow & & \downarrow{\scriptstyle F\psi} \\ \partial'X' & \xrightarrow{f'} & FY' \end{array}$$

commutes. Then $\mathcal{P}$ has finite sums and a small subcategory $\mathcal{P}_0(X \in \mathcal{C}'_0,\ Y \in \mathcal{C}_0)$ such that each object of $\mathcal{P}$ is isomorphic to an object of $\mathcal{P}_0$.

Let $\mathcal{S}$ be the collection of pairs $((X,Y,f),(X',Y',f'))$ of objects of $\mathcal{P}$ for which $Y \cong Y'$. Let $(x,x') \sim (y,y')$ if there are objects u and v of $\mathcal{P}$ such that $x + u \cong y + v$ and $x' + u \cong y' + v$. Then the set of equivalence classes $\mathcal{S}/\sim$ forms an abelian group under the operation induced by the sum.

One has a homomorphism $\beta : K(\mathcal{C}'_{cl}) \longrightarrow \mathcal{S}/\sim$, where $\mathcal{C}'_{cl}$ is the subcategory of closed objects of $\mathcal{C}'$ by sending (X,X') into $((X,\emptyset,j),(X',\emptyset,j'))$ where $\emptyset$ is an initial object of $\mathcal{C}$ and j,j' are the unique isomorphisms of initial objects.

If one has a homomorphism

$$\alpha : \mathcal{S}/\sim \longrightarrow K(\mathcal{C}'_{cl})/(\partial'_*K(\mathcal{C}') + F_*K(\mathcal{C}_{cl}))$$

such that the composition with β is the quotient homomorphism of $K(\mathcal{C}'_{cl})$, then one can define a relative cobordism semigroup as follows:

For objects (X,Y,f) and (X',Y',f') of $\mathcal{P}$, one writes $(X,Y,f) \equiv (X',Y',f')$ if there exist objects U and U' of $\mathcal{C}$ with $Y + \partial U \cong Y' + \partial U'$ and for which $\alpha((X+FU,Y+\partial U,f+tU),(X'+FU',Y'+\partial U',f'+tU')) = 0$. Using the fact that α is a homomorphism one easily sees that $\equiv$ is an equivalence relation. The relative cobordism semigroup $\Omega(F,t,\alpha)$ is the set of equivalence classes under $\equiv$ of elements of $\mathcal{P}$ with the sum induced by the sum in $\mathcal{P}$.

One has homomorphisms

$$\partial : \Omega(F,t,\alpha) \longrightarrow \Omega(\mathcal{C},\partial,i) : (X,Y,f) \longrightarrow Y,$$

$$F_* : \Omega(\mathcal{C},\partial,i) \longrightarrow \Omega(\mathcal{C}',\partial',i') : Y \longrightarrow FY, \text{ and}$$

$$i : \Omega(\mathcal{C}',\partial',i') \longrightarrow \Omega(F,t,\alpha) : X \longrightarrow (X,\emptyset,j)$$

and the triangle

$$\begin{array}{ccc} \Omega(\mathcal{C},\partial,i) & \xrightarrow{F_*} & \Omega(\mathcal{C}',\partial',i') \\ & {\scriptstyle\partial}\nwarrow \quad \swarrow{\scriptstyle i} & \\ & \Omega(F,t,\alpha) & \end{array}$$

is easily seen to have period 2 (i.e. $\partial i = iF_* = F_*\partial = 0$).

In order to clarify the relationship between the homomorphism α and the joining of two manifolds along their common boundary, consider elements (X,Y,f) of $\mathcal{P}$ as a manifold with boundary together with additional structure on its boundary. For $((X,Y,f), (X',Y',f')) \in \mathcal{S}$ choose an isomorphism $g : Y \xrightarrow{\cong} Y'$ and let $\alpha(x,x')$ be the class of $X \cup_k (-X')$, where $-X'$ is X' with its opposite structure (e.g. orientation), and the boundaries of X and X' are identified via $k = f'^{-1}F(g)f$. This class does not depend on the choice of g, for if g' is another isomorphism, one may attach $X' \times I$ to $(X \cup_k (-X')) \times I \cup [\pm(X \cup_{k'} (-X'))] \times I$ so that the difference of two representatives is cobordant to $X \cup_{k''} (-X)$, where $k'' = f^{-1}F(g^{-1}g')f$. Identifying $\partial X \times 0$ with $\partial X \times 1$ using k'' in $X \times I$ gives a cobordism of $X \cup_{k''} (-X)$ and $\partial X \times I$ with ends identified by k'' - but this is isomorphic via f to the image under F of $Y \times I$ with ends identified using $g^{-1}g'$. Thus α does not depend on the choice of g.

With this choice of α, suppose one has $(X,Y,f) \equiv (X',Y',f')$. One may then find a cobordism of Y and Y', say $\partial V = Y - Y'$ so that $X \cup (-V) \cup (-X')$ is cobordant to a closed manifold D with additional structure. Thus one may find a cobordism of Y and Y', $U = V + D$, $\partial U = Y - Y'$, so that $X \cup (-U) \cup (-X')$ bounds. This is the usual geometric description for cobordism of manifolds with boundary.

Remark: One may let $\mathcal{C}$ be the subcategory of $\mathcal{C}'$ consisting of initial objects, with F the inclusion. Then β is epic, uniquely determining α. The relative cobordism semigroup in this case is then identifiable with the cobordism semigroup of $\mathcal{C}'$.

Chapter II

Manifolds With Structure - The Pontrjagin-Thom Theorem

The standard cobordism theories are based on manifolds with additional structure on the tangent or normal bundle. The exposition given here is taken from the paper: R. K. Lashof: Poincaré duality and cobordism, Trans. Amer. Math. Soc., 109 (1963), 257-277.

Denote by $G_{r,n}$ the Grassmann manifold of unoriented r-planes in R^{r+n} and let γ^r_n be the r-plane bundle over $G_{r,n}$ consisting of pairs: an r-plane in R^{n+r} and a point in that r-plane. Then $BO_r = \lim_{n\to\infty} G_{r,n}$ with universal r-plane bundle $\gamma^r = \lim_{n\to\infty} \gamma^r_n$.

Definition: Let $f_n : B_n \longrightarrow BO_n$ be a fibration. If ξ is an n-dimensional vector bundle over the space X classified by the map $\xi : X \longrightarrow BO_n$, then a (B_n,f_n) structure on ξ is a homotopy class of liftings to B_n of the map ξ; i.e. an equivalence class of maps $\hat{\xi} : X \longrightarrow B_n$ with $f_n \circ \hat{\xi} = \xi$, where $\hat{\xi}$ and $\tilde{\xi}$ are equivalent if they are homotopic by a homotopy $H : X \times I \longrightarrow B_n$ such that $f_n \circ H(x,t) = \xi(x)$ for all $(x,t) \in X \times I$.

Note: A (B_n,f_n) structure depends on the specific map into BO_n. There is no way to make (B_n,f_n) structures correspond for equivalent bundles, since the correspondence is dependent upon the choice of the equivalence.

Let M^n be a compact differentiable (C^∞) manifold (with or without boundary) and let $i : M^n \longrightarrow R^{n+r}$ be an imbedding. The normal bundle of i is the quotient of the pullback of the tangent bundle of R^{n+r}, $i^*\tau(R^{n+r})$, by the subbundle $\tau(M)$. Giving $\tau(R^{n+r}) = R^{n+r} \times R^{n+r}$ the Riemannian metric obtained from the usual inner product in Euclidean space, the total

space N of the normal bundle may be identified with the orthogonal complement of $\tau(M)$ in $i^*\tau(R^{n+r})$, or the fiber of N at m may be identified with the subspace of $R^{n+r} \times R^{n+r}$ consisting of vectors (m,x) with x orthogonal to $i_*\tau(M)_m$. The normal map of i is given by sending m into $N_m \in G_{r,n}$, covered by the bundle map $n : N \longrightarrow \gamma^r_n : (m,x) \longrightarrow (N_m,x)$. Composing with the inclusion into γ^r provides a map $\nu(i) : M \longrightarrow BO_r$ which classifies the normal bundle of the imbedding i.

Lemma: If r is sufficiently large (depending only on n), there is a one-to-one correspondence between the (B_r,f_r) structures for the normal bundles of any two imbeddings $i_1,i_2 : M^n \longrightarrow R^{n+r}$.

Proof: For r sufficiently large, any two imbeddings i_1, i_2 of M^n in R^{n+r} are regularly homotopic and any two such regular homotopies are homotopic through regular homotopies leaving endpoints fixed. (A regular homotopy is a homotopy $H : M \times I \longrightarrow R^{n+r}$ such that each $H(\ ,t)$ is an immersion and such that the differentials $H(\ ,t)_* : \tau(M) \longrightarrow \tau(R^{n+r})$ define a homotopy). See M. Hirsch: Immersions of manifolds, Trans. Amer. Math. Soc., 93 (1959), 242-276. Then a regular homotopy from i_1 to i_2 gives a homotopy from $\nu(i_1)$ to $\nu(i_2)$, and two homotopies defined in this way are themselves homotopic relative to endpoints. Thus one has a well-defined equivalence for the two bundles. Applying the homotopy lifting property for the map f_r then establishes the correspondence quite easily. **

Definition: Suppose one is given a sequence (B,f) of fibrations $f_r : B_r \longrightarrow BO_r$ and maps $g_r : B_r \longrightarrow B_{r+1}$ such that the diagram

$$\begin{array}{ccc} B_r & \xrightarrow{g_r} & B_{r+1} \\ \downarrow{\scriptstyle f_r} & & \downarrow{\scriptstyle f_{r+1}} \\ BO_r & \xrightarrow{j_r} & BO_{r+1} \end{array}$$

commutes, j_r being the usual inclusion. A (B_r,f_r) structure on the normal bundle of M^n in R^{n+r} defines a unique (B_{r+1},f_{r+1}) structure via the inclusion $R^{n+r} \subset R^{n+r+1}$. A (B,f) structure on M^n is an equivalence class of sequences of (B_r,f_r) structures $\xi = (\xi_r)$ on the normal bundle of M, two such sequences being equivalent if they agree for sufficiently large r. A (B,f) manifold is a pair consisting of a manifold M^n and a (B,f) structure on M^n.

If W^w is a manifold and M^m is a submanifold of W with trivialized normal bundle, one may imbed M in R^{m+r}, r large, and extend by means of the trivialization to an imbedding of a neighborhood of M in W into $R^{w+r} = R^{m+r} \times R^{w-m}$ so that the neighborhood meets R^{m+r} orthogonally along M. This may then be extended to an imbedding of W in R^{w+r}. The normal planes to M in R^{m+r} are then the restriction to M of the normal planes to W in R^{w+r}. If $\tilde{\nu} : W \longrightarrow B_r$ is a lifting of the normal map, then $\tilde{\nu}|_M$ is a lifting for the normal map of M. Thus a (B,f) structure on W induces a well-defined (B,f) structure on M.

Remarks: 1) The induced (B,f) structure depends only on the equivalence class of the trivialization, not on the specific choice of trivialization.

2) If $f : M \longrightarrow W$ is an isomorphism of manifolds, the normal bundle is trivialized, being zero dimensional. If $i : M \longrightarrow W$ is the inclusion

of the boundary, there are two choices of trivialization, via the choice of inner or outer normal. If $j : M \longrightarrow W$ is the inclusion of a direct summand, then the normal bundle is again zero dimensional, so trivialized.

Definition: The cobordism category of (B,f) manifolds is the category whose objects are compact differentiable manifolds with (B,f) structure and whose maps are the boundary preserving differentiable imbeddings with trivialized normal bundle such that the (B,f) structure induced by the map coincides with the (B,f) structure on the domain manifold. The functor ∂ applied to a (B,f) manifold W is the manifold ∂W with (B,f) structure induced by the inner normal trivialization, and ∂ on maps is restriction. The natural transformation i is the inclusion of the boundary with inner normal trivialization.

The cobordism semigroup of this category will be denoted $\Omega(B,f)$. The sub-semigroup of equivalence classes of n-dimensional closed manifolds will be denoted $\Omega_n(B,f)$. Clearly $\Omega(B,f)$ is the direct sum of the $\Omega_n(B,f)$.

Proposition: The cobordism semigroup $\Omega(B,f)$ is an abelian group.

Proof: Let M^n be a closed manifold imbedded in R^{n+r} for some large r with $\tilde{\nu} : M \longrightarrow B_r$ a lifting of the normal map. Extend to an imbedding of $M \times I$ in $R^{n+r} \times R = R^{n+r+1}$ by the usual inclusion of I in R. The normal map for $M \times I$ is the composition of the projection on M and the normal map of M. Thus the lifting for M defines a (B,f) structure on $M \times I$ which induces the given structure on $M \times 0$. The inner normal along $M \times 1$ gives rise to an induced structure on $M \times 1$, and with these structures one has $M + M \times 1 = \partial(M \times I)$ in the category. Thus the structure on $M \times 1$ is an inverse for the structure on M in $\Omega(B,f)$. **

Considering BO_r as the space of r-planes contained in some finite dimensional subspace R^s of R^∞ and taking the usual inner product on the subspace of R^∞ consisting of vectors with only finitely many non-zero components, one obtains a Riemannian metric on the universal bundle γ^r. If ξ is an r-plane bundle over a space X classified by the map $\xi : X \longrightarrow BO_r$, one has induced a Riemannian metric on ξ. (Note: For the normal bundle of a manifold this coincides with the metric obtained from the splitting). The Thom space of ξ, $T\xi$, is the space obtained from the total space of ξ by collapsing all vectors of length at least one to a point, denoted ∞. If ξ is the bundle induced from a bundle $\eta : Y \longrightarrow BO_r$, by a map $g : X \longrightarrow Y$, then the usual bundle map $\xi = g^*\eta \longrightarrow \eta$ induces a map $Tg : T\xi \longrightarrow T\eta$.

The map $j_r : BO_r \longrightarrow BO_{r+1}$ induces a vector bundle $j_r^*(\gamma^{r+1})$ over BO_r which may be identified as the Whitney sum of γ^r and a trivial line bundle. Then $Tj_r^*(\gamma^{r+1})$ may be identified as the suspension of $T\gamma^r$.

One then has a commutative diagram

$$\begin{array}{ccc} \Sigma TB_r & \xrightarrow{Tg_r} & TB_{r+1} \\ \downarrow{\scriptstyle \Sigma Tf_r} & & \downarrow{\scriptstyle Tf_{r+1}} \\ \Sigma TBO_r & \xrightarrow{Tj_r} & TBO_{r+1} \end{array}$$

and a homomorphism $Tg_r \circ \Sigma : \pi_{n+r}(TB_r,\infty) \longrightarrow \pi_{n+r+1}(TB_{r+1},\infty)$ of the homotopy groups, where Σ denotes suspension, and TBO_r, TB_r denote the Thom spaces $T\gamma^r$ and $Tf_r^*(\gamma^r)$.

The main theorem is the generalized Pontrjagin-Thom theorem:

Theorem: The cobordism group of n-dimensional (B,f) manifolds $\Omega_n(B,f)$ is isomorphic to $\lim_{r\to\infty} \pi_{n+r}(TB_r,\infty)$.

Proof:

A) Definition of the homomorphism $\Theta : \Omega_n(B,f) \longrightarrow \lim_{r\to\infty} \pi_{n+r}(TB_r,\infty)$.

Let $\sigma \in \Omega_n(B,f)$ be represented by a (B,f) manifold M^n. Let $i : M \longrightarrow R^{n+r}$ be an imbedding with a lifting $\tilde{\nu} : M \longrightarrow B_r$ which defines the given (B,f) structure on M. Let N denote the total space of the normal bundle of M, thought of as a subspace of $R^{n+r} \times R^{n+r}$. Under the evaluation map $e : R^{n+r} \times R^{n+r} \longrightarrow R^{n+r} : (a,b) \longrightarrow a+b$, the subspace N is mapped differentiably and on $M = M \times 0 \subset N$ this map restricts to the imbedding i. For some sufficiently small $\varepsilon > 0$, the subspace of N consisting of vectors of length less than or equal to ε, N_ε, is imbedded by this map $e|_N$.

To define a map $S^{n+r} \longrightarrow Tf_r^*(\gamma^r)$, begin by considering S^{n+r} as $R^{n+r} \cup \infty$ and let $c : S^{n+r} \longrightarrow N_\varepsilon/\partial N_\varepsilon$ by collapsing all points of S^{n+r} outside or on the boundary of N_ε to a point. Multiplication by $1/\varepsilon$ induces a map $N_\varepsilon/\partial N_\varepsilon \longrightarrow TN$, denoted by ε^{-1}. The map $\bar{n} \times (\tilde{\nu}\circ\pi) : N \to \gamma^r \times B_r$, where $\bar{n}$ is the composition of n with the inclusion of γ_n^r in γ^r and π is the projection of N on M, is a bundle map into $f_r^*(\gamma^r)$ and induces the map $T(\bar{n} \times (\tilde{\nu}\circ\pi)) : TN \longrightarrow TB_r$. The composition $\theta = T(\bar{n} \times (\tilde{\nu}\circ\pi)) \circ \varepsilon^{-1} \circ c$ is a map of pairs $(S^{n+r},\infty) \longrightarrow (TB_r,\infty)$.

Replacing ε by a smaller value does not change the homotopy class of θ since the maps $\varepsilon^{-1}\circ c$ will be homotopic. Replacing $\tilde{\nu}$ by an equivalent lifting simply gives a homotopy of $T(\bar{n} \times (\tilde{\nu}\circ\pi))$ and so does not change the homotopy class of θ. Clearly, the map $M \xrightarrow{i} R^{n+r} \subset R^{n+r+1}$ gives rise to $Tg_r \circ \Sigma\theta$ and thus one has defined an element of $\lim_{r\to\infty} \pi_{n+r}(TB_r,\infty)$ represented by the map θ.

To show that this element depends only on the cobordism class of the manifold M and not on the choice of the imbedding, let W be a (B,f) manifold and $j : M + \partial W \longrightarrow R^{n+r}$ an imbedding with a lift $\bar{\nu} : M + \partial W \longrightarrow B_r$ giving the same (B_r,f_r) structure on M (here r is assumed large). Let $H : M \times I \longrightarrow R^{n+r}$ be a regular homotopy of the imbeddings i and $j|_M$ chosen so that $H(x,t)$ is $i(x)$ if $t < \delta_1$ and is $j(x)$ if $t > 1-\delta_2$ and let $k : W \longrightarrow R^{n+r} \times (0,1]$ be a map agreeing with $j \times 1$ on ∂W and imbedding a tubular neighborhood of ∂W orthogonally along $j(\partial W) \times 1$. The map $(H \times \pi_2) + k : M \times I + W \longrightarrow R^{n+r} \times I$ is an imbedding on a closed neighborhood of the boundary and may be homotoped to an imbedding $F : M \times I + W \longrightarrow R^{n+r} \times I$ by a homotopy fixed on that neighborhood of the boundary. $F|_{M\times I}$ is a regular homotopy and corresponding to its normal map one may find a covering map $M \times I \longrightarrow B_r$ agreeing with $\tilde{\nu}$ on $M \times 0$. Since the normal map is constant near $M \times 1$, one may modify the lift to agree with $\bar{\nu}$ on $M \times 1$. Since the (B,f) structure on ∂W is induced from that of W one may find a lift of W agreeing with $\bar{\nu}$ on ∂W. Following the previous construction with the imbedding $F : M \times I + W \longrightarrow R^{n+r} \times I$, one has a collapse $S^{n+r} \times I \longrightarrow N_\varepsilon/\partial N_\varepsilon$ where N_ε is a neighborhood of (image F), a map $\varepsilon^{-1} : N_\varepsilon/\partial N_\varepsilon \longrightarrow TN$ and the map $T(\bar{n} \times (\tilde{\nu}\circ\pi)) : TN \longrightarrow TB_r$ which compose to give a map $S^{n+r} \times I \longrightarrow TB_r$. This provides a homotopy for the maps θ defined by i and j.

Taking W empty shows that the class of θ is independent of the imbedding of M. Further, if $M \equiv M'$ with $M + \partial W \tilde{=} M' + \partial W'$, then $\theta(M) - \theta(M+\partial W) = \theta(M'+\partial W') - \theta(M')$ so that the class of the map θ depends only on the cobordism class of M.

B) θ is a homomorphism.

If M_1 and M_2 represent two classes in $\Omega_n(B,f)$, choose imbeddings $i_\alpha : M_\alpha \longrightarrow R^{n+r}$ for which the last coordinate is positive for i_1 and negative for i_2. If tubular neighborhoods are chosen small enough to lie in the same half spaces then $\theta([M_1] + [M_2])$ is represented by $S^{n+r} \xrightarrow{d} S^{n+r} \vee S^{n+r} \xrightarrow{\theta_1 \vee \theta_2} TB_r$, where d collapses the equator of S^{n+r} and θ_α represents $\theta([M_\alpha])$. Since this map represents the sum of the homotopy classes, θ is a homomorphism.

C) θ is epic.

Let $\theta : (S^{n+r},p) \longrightarrow (TB_r,\infty)$, r large, represent a class in $\lim_{r\to\infty} \pi_{n+r}(TB_r,\infty)$. Then $Tf_r\circ\theta : (S^{n+r},p) \longrightarrow (TBO_r,\infty)$, and since $TBO_r = T\gamma^r = \lim_{s\to\infty} T\gamma^r_s$ with S^{n+r} compact, $Tf_r\circ\theta(S^{n+r}) \subset T\gamma^r_s$ for some s. The map $Tf_r\circ\theta$ may be deformed to a map h_r so that

1) h_r is differentiable on the preimage of some open set of $T\gamma^r_s$ containing $G_{r,s}$ and is transverse regular on $G_{r,s}$. Note: $T\gamma^r_s - \infty$ is a differentiable manifold.

2) If $M^n = h_r^{-1}(G_{r,s})$, h_r is a bundle map on a normal tube of M^n in $R^{n+r} = S^{n+r}-p$.

3) The map $Tf_r\circ\theta$ agrees with h_r on the preimage V of a closed neighborhood of ∞.

Since $h_r|_M$ classifies the normal bundle of M, one may assume (by a further homotopy if necessary) that $h_r|_M$ is the normal map $\nu : M \longrightarrow G_{r,n} \subset G_{r,s}$ and that h_r is given by the usual translation of vectors to the origin on a normal tubular neighborhood of M.

Now $Tf_r : TB_r \longrightarrow TBO_r$ is a fibration except at the point ∞, and since $Tf_r \circ \theta(S^{n+r} - \text{interior } V)$ does not contain ∞, the covering homotopy theorem applies so that the deformation of $Tf_r \circ \theta$ to h_r on $S^{n+r} - \text{interior } V$ may be covered by a homotopy of θ on $S^{n+r} - \text{interior } V$ which is pointwise fixed on the boundary of V. Taking the homotopy to be constant on V, one may cover the homotopy of $Tf_r \circ \theta$ to h_r by a homotopy of θ to a new map θ_1. The inverse image of B_r under θ_1 is the inverse image of BO_r under h_r, which is M. Further $\theta_1|_M$ is a lift of the normal map $h_r|_M$.

Thus one has a (B_r, f_r) structure on M in R^{n+r} and hence a (B,f) structure on M. Using the given imbedding of M in R^{n+r} with the lift $\theta_1|_M$, the resulting map θ_2 agrees with θ_1 on a neighborhood N_ε of M and since $TB_r - B_r$ deforms to ∞, one may homotope θ_1 to agree with θ_2 by pushing the complement of N_ε to ∞. Then $\theta([M])$ is the class of θ.

D) θ is monic.

Let M be a (B,f) manifold such that $\theta([M]) = 0$. Thus for some large r the standard map $\theta_0 : S^{n+r} \longrightarrow TB_r$ defined by M is homotopic to the trivial map $\theta_1 : S^{n+r} \longrightarrow \infty$. One may choose the homotopy $L : S^{n+r} \times I \longrightarrow TB_r$ so that $L_t = \theta_0$ for $t \in [0,\eta]$. By compactness $Tf_r \circ L(S^{n+r} \times I) \subset T\gamma^r_s$ for some s $(\geq n)$. As above, one may homotope $Tf_r \circ L$ (relative to $N_\varepsilon(M) \times [0,\eta]$) in a neighborhood of $G_{r,s}$ to a map H_r which is differentiable near and transverse on $G_{r,s}$. $W = H_r^{-1}(G_{r,s})$ is a submanifold of $R^{n+r} \times I$ with $\partial W = M$ meeting $R^{n+r} \times 0$ orthogonally along M. One may also assume $H_r|_W$ is the normal map and H_r agrees with the usual translation of vectors map on

a neighborhood of W. Applying covering homotopy, one may deform L to a map $\theta : S^{n+r} \times I \longrightarrow TB_r$ with $\theta_t = \theta_0$ for small t, $\theta_1 = \theta|S^{n+r} \times 1$ with $\theta|_W$ covering the normal map H_r of W. This defines a (B,f) structure on W which induces the original (B,f) structure on M. Thus $M + \partial\emptyset = \emptyset + \partial W$ and $[M]$ is the zero class of $\Omega_n(B,f)$. **

Tangential Structures

It is frequently desirable to define (B,f) structures on manifolds by means of structures on the stable tangent bundle. Let $B = \lim (B_r, g_r)$ and $BO = \lim (BO_r, j_r)$, with $f = \lim f_r : B \longrightarrow BO$. The map $I_{n,N} : G_{n,N} \longrightarrow G_{N,n}$ obtained by assigning to each n plane its orthogonal N plane induces a map $I : BO \longrightarrow BO$, with I^2 the identity. $f : B \longrightarrow BO$ is a fibration and one has the induced fibration $f^* : B^* = I^*B \longrightarrow BO$. Since $I^2 = 1$, I^*B^* is again B. The induced bundle maps give a diagram

$$\begin{array}{ccccc} B & \xrightarrow{I'} & B^* & \xrightarrow{I^*} & B \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f^*} & & \downarrow{\scriptstyle f} \\ BO & \xrightarrow{I} & BO & \xrightarrow{I} & BO \end{array}$$

with I^*I' and $I'I^*$ both being identity maps.

If M^n is imbedded in R^{n+N}, N large, the maps $\nu_N : M^n \longrightarrow G_{N,n}$ and $\tau_N : M^n \longrightarrow G_{n,N}$ obtained by translation of normal and tangent vectors are related by $\tau_N = I_{N,n}\nu_N$. Following these by the inclusions one has maps $\tau : M \longrightarrow BO$ and $\nu : M \longrightarrow BO$ with $\tau = I\nu$.

A (B,f) structure on M as previously defined is precisely a homotopy class (through liftings) of liftings to B of the map $\nu : M \longrightarrow BO$. The maps I' and I^* define an obvious equivalence between these classes of liftings of ν and the homotopy classes (through liftings) of liftings to B^* of $\tau : M \longrightarrow BO$. Such a class of liftings of τ is a (B^*,f^*) structure on the stable tangent bundle of M.

Structures For Sequences Of Maps

If instead of fibrations one is given only spaces C_r and maps $f_r : C_r \longrightarrow BO_r$, $g_r : C_r \longrightarrow C_{r+1}$ such that $f_{r+1}g_r$ is homotopic to $j_r f_r$ one may replace the maps (C_r,f_r) by homotopy equivalent fibrations. The resulting maps g_r may be deformed inductively to give commutative diagrams by means of covering homotopy. A (C,f) structure is then a structure for some such fibration sequence (chosen). Since the cobordism group is given by homotopy of the Thom complex, which depends only on the homotopy type of the fibrations, the resulting cobordism group does not depend on the choice of equivalent fibration sequence.

Ring Structure

If one has an r plane in $r+s$ space and an r' plane in $r'+s'$ space, they span an $r+r'$ plane in $R^{r+r'+s+s'} = R^{r+s} \times R^{r'+s'}$. This defines a map $G_{r,s} \times G_{r',s'} \longrightarrow G_{r+r',s+s'}$ and induces a map $BO_r \times BO_{r'} \longrightarrow BO_{r+r'}$, corresponding to the Whitney sum of vector bundles. $G_{0,0}$ is a point and provides a base point in each $G_{r,s}$ (the usual $R^r \subset R^{r+s}$) so that $BO_r \vee BO_{r'}$ is mapped via standard inclusions.

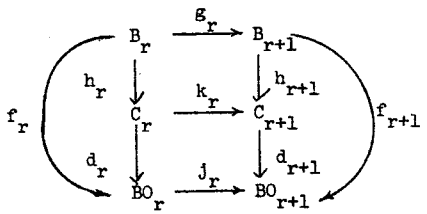
The twisted map $BO_{r'} \times BO_r \longrightarrow BO_r \times BO'_r \longrightarrow BO_{r+r'}$ is homotopic to the usual map $BO_{r'} \times BO_r \longrightarrow BO_{r+r'}$ by a rotation of $R^{r+r'+s+s'}$ to interchange factors. This gives the usual homotopy commutative H-space structure on BO.

Having similar multiplications $B_r \times B_s \longrightarrow B_{r+s}$ so that the maps f_r preserve products up to homotopy, one may define a ring structure in (B,f) cobordism, for the multiplication defines a (B,f) structure on the product manifold $M^n \times M'^{n'} \subset R^{n+N} \times R^{n'+N'} = R^{n+n'+N+N'}$.

The map $B_r \times B_s \longrightarrow B_{r+s}$ induces a map $TB_r \wedge TB_s \longrightarrow TB_{r+s}$ giving a product in the stable homotopy, making it into a ring. This ring structure is the same as that of the cobordism groups.

Relative Groups

If one has commutative diagrams



in which h_r and d_r are fibrations and g_r and k_r are fibre preserving maps, a (B,f) structure induces a (C,d) structure by means of the 'reduction' h. This gives a functor h from the cobordism category of (B,f) manifolds to that of (C,d) manifolds. An $(n+1)$ dimensional manifold W with boundary having a (B,f) structure on its boundary inducing via h the same (C,d) structure on ∂W as is induced by a (C,d)

structure on W is a relative manifold. Using the standard 'piecing together' homomorphism one has defined a relative cobordism semigroup.

If one imbeds ∂W in R^{n+r} (r large) with lifting of the normal map to B_r, extends to an imbedding of W (orthogonally along the boundary using a tubular neighborhood) in $R^{n+r} \times [0,\infty) = H^{n+r+1}$, selecting a lifting of the normal map of W to C_r which agrees with the h-induced lifting on ∂W, then one may apply the tubular neighborhood map to this imbedding-lifting situation to construct a map

$$\theta : (D^{n+r+1}, S^{n+r}, \infty) = (H^{n+r+1} \cup \infty, R^{n+r} \cup \infty, \infty) \longrightarrow (TC_r, TB_r, \infty).$$

If W is cobordant to W' (relatively) one may find a (B,f) manifold U giving a cobordism of ∂W and $\partial W'$ so that with proper boundary identifications the closed (C,d) manifold $W \cup (-U) \cup (-W')$ bounds. One may imbed U in $R^{n+r} \times I$ to give the proper identifications at $\partial W \times 0$ and $\partial W' \times 1$, with a lifting of the normal map to B_r and fill in the manifold with (C,d) structure in $H^{n+r+1} \times I$ along its boundary $W \cup (-U) \cup (-W') \subset H^{n+r+1} \times 0 \cup R^{n+r} \times I \cup H^{n+r+1} \times 1$. Ignoring corners (which don't affect the homotopy situation, but rather involve the identification of $D^{n+r+1} \times I$) the normal maps and their liftings define a homotopy

$$L : (D^{n+r+1} \times I, S^{n+r} \times I, \infty \times I) \longrightarrow (TC_r, TB_r, \infty)$$

of the maps for W and W'.

Ignoring lots of details one sees easily that the $(n+1)$ dimensional relative cobordism group ($-W$ being constructed from $W \times I$) is isomorphic to the stable homotopy group $\lim_{r\to\infty} \pi_{n+r+1}(TC_r, TB_r, \infty)$. Further, the cobordism triangle is identifiable as the exact homotopy sequence of the 'pair' (TC,TB).

Chapter III

Characteristic Classes and Numbers

As mentioned in the introduction, the determination of invariants which distinguish manifolds in one of the principal aims of differential topology. In the framework of cobordism theory, the use of characteristic classes provides invariants called characteristic numbers which are cobordism invariants. In order to set up the machinery for these invariants, the ideas of generalized cohomology theory play a central role, and for this the basic reference is G. W. Whitehead: Generalized homology theories, Trans. Amer. Math. Soc., 102 (1962), 227-283.

Definition: A spectrum $\underset{\sim}{E}$ is a sequence $\{E_n | n \in Z\}$ of spaces with base point together with a sequence of maps $e_n : \Sigma E_n \longrightarrow E_{n+1}$, Σ being the suspension. If $\underset{\sim}{F} = \{F_n, f_n\}$ is another spectrum, a map h from $\underset{\sim}{E}$ to $\underset{\sim}{F}$ is a sequence of maps $h_n : E_n \longrightarrow F_n$ with $h_{n+1} \circ e_n = f_n \circ \Sigma h_n$.

Examples: 1) The sphere spectrum $\underset{\sim}{S} = \{S^n, \sigma_n\}$ where $\sigma_n : \Sigma S^n \longrightarrow S^{n+1}$ is the identity map.

2) If (B,f) is a sequence of fibrations $f_r : B_r \longrightarrow BO_r$ with maps $g_r : B_r \longrightarrow B_{r+1}$ as in Chapter II, then $\underset{\sim}{TB} = \{TB_r, Tg_r\}$ is a spectrum, known as the Thom spectrum of the family (B,f). In particular, the maps $Tf_r : TB_r \longrightarrow TBO_r$ define a map of spectra $Tf : \underset{\sim}{TB} \longrightarrow \underset{\sim}{TBO} = \{TBO_r, Tj_r\}$.

If one chooses base points $b_r \in B_r$ such that $g_r b_r = b_{r+1}$, then the bundle $f_r^*(\gamma^r)$ induces a trivial r-plane bundle δ^r over b^r, defining a map $Tb_r : T\delta^r \longrightarrow TB_r$. Since $T\delta^r = S^r$, this gives a map of spectra $Tb : \underset{\sim}{S} \longrightarrow \underset{\sim}{TB}$. Note: The identification of $T\delta^r$ with S^r requires a choice of framing of the fiber over b_r.

Definition: The homology and cohomology groups with coefficients in the spectrum $\underset{\sim}{E}$ are defined by

$$H_n(X,A;\underset{\sim}{E}) = \lim_{i\to\infty} \pi_{n+i}((X/A)\wedge E_i) \quad \text{and}$$

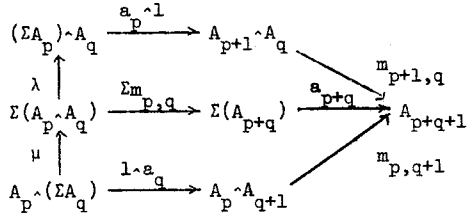
$$H^n(X,A;\underset{\sim}{E}) = \lim_{i\to\infty} [\Sigma^i(X/A), E_{n+i}]$$

where X/A is the space obtained from X by collapsing A to a point (the base point), $\wedge$ is the smash product $U\wedge V = U \times V/(U \times *) \cup (* \times V)$, and $[\ ,\ ]$ denotes homotopy classes of maps.

$H^*(\ ;\underset{\sim}{E})$ and $H_*(\ ;\underset{\sim}{E})$ are functors satisfying all the axioms of Eilenberg-Steenrod as cohomology and homology theories with the exception of the dimension axiom.

One defines $H_*(X;\underset{\sim}{E})$ to be $H_*(X,\emptyset;\underset{\sim}{E})$ where $\emptyset$ is the empty set and $X/\emptyset$ is the disjoint union of X and a point. If Y is a space with base point p, one writes $\tilde{H}_*(Y;\underset{\sim}{E})$ for $H_*(Y,p;\underset{\sim}{E})$.

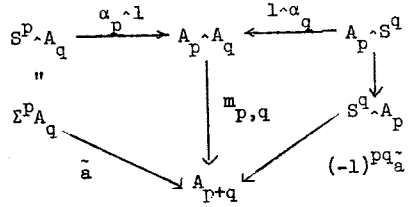
Definition: A ring spectrum is a spectrum $\underset{\sim}{A} = \{A_p, a_p\}$ with a map $\alpha : \underset{\sim}{S} \longrightarrow \underset{\sim}{A}$ and a pairing $m : (\underset{\sim}{A},\underset{\sim}{A}) \longrightarrow \underset{\sim}{A}$; i.e. a collection of maps $m_{p,q} : A_p\wedge A_q \longrightarrow A_{p+q}$ such that the maps of the diagram



represent classes of the group $[\Sigma(A_p\wedge A_q), A_{p+q+1}]$ related by

$$[m_{p+1,q}\circ(a_p\wedge 1)\circ\lambda] = [a_{p+q}\circ\Sigma m_{p,q}] = (-1)^p[m_{p,q+1}\circ(1\wedge a_q)\circ\mu];$$

such that the diagram



commutes, where $\tilde{a}$ is the multiple composition of the suspensions of maps a_j and $(-1)^{pq}\tilde{a}$ is a map whose class in the group $[\Sigma^q A_p, A_{p+q}]$ is $(-1)^{pq}$ times the class of $\tilde{a}$. [Note: If $q = 0$ this is not a group, but $(-1)^{pq} = 1$ so that one does not need the group structure to find a map.]

Example: Let R be a ring with unit, $K(R,n)$ an Eilenberg-MacLane space (the only non-zero homotopy group being R in dimension n) and let $\kappa_n : \Sigma K(R,n) \longrightarrow K(R,n+1)$ be a map corresponding to the identification $K(R,n) \longrightarrow \Omega K(R,n+1)$. The spectrum $\underset{\sim}{K}(R) = \{K(R,n),\kappa_n\}$ is a ring spectrum and $H^n(X;\underset{\sim}{K}(R))$ is the usual cohomology with coefficients in R.

With a ring spectrum one has the usual sorts of products, such as cup products in cohomology making $H^*(X;\underset{\sim}{A})$ into a commutative ring with unit.

Definition: Let (B,f) be a sequence of fibrations $f_r : B_r \longrightarrow BO_r$. A Thom class is a map of spectra $U : \underset{\sim}{TB} \longrightarrow \underset{\sim}{A}$, where $\underset{\sim}{A}$ is a ring spectrum.

Now let M^n be a (B,f) manifold, $i : M^n \longrightarrow H^{n+r}$ an imbedding with ∂M imbedded in R^{n+r-1} with the usual orthogonal framing along a tubular neighborhood of $\partial M = \partial M \times 0$ (denoting the neighborhood by $\partial M \times [0,1]$). Let N be the normal bundle of M and N' the normal bundle of ∂M. Let $\nu : M \longrightarrow B_r$ be a lifting defining the (B,f) structure on M with $T\nu : TN \longrightarrow TB_r$ the induced map on the Thom complex.

Consider the map

$$N \xrightarrow{\Delta} N \times N \xrightarrow{\pi \times p} M \times TN \longrightarrow (M/\partial M) \wedge TN$$

where Δ is the diagonal map, π the bundle projection, p is the collapse onto the Thom complex, and the last map is the obvious collapse. Under this map the vectors of norm at least one are sent to the base point, as are all vectors over ∂M, i.e. N'. Thus one induces a map $\phi : TN/TN' \longrightarrow (M/\partial M) \wedge TN$.

Letting $c : H^{n+r} \longrightarrow TN$ be the standard (scaled) collapse, the projection into TN/TN' sends R^{n+r-1} into the base point also and hence defines a collapse $c : S^{n+r} = (H^{n+r} \cup \infty)/(R^{n+r-1} \cup \infty) \longrightarrow TN/TN'$.

Letting $U = \{U_r\} : \underset{\sim}{TB} \longrightarrow \underset{\sim}{A}$ be a Thom class, one has a composite map

$$S^{n+r} \xrightarrow{c} TN/TN' \xrightarrow{\phi} (M/\partial M) \wedge TN \xrightarrow{1 \wedge T\nu} (M/\partial M) \wedge TB_r \xrightarrow{1 \wedge U_r} (M/\partial M) \wedge A_r$$

which represents an element of $\pi_{n+r}((M/\partial M) \wedge A_r)$. Letting r go to infinity defines a class $[M,\partial M] \in H_n(M,\partial M;\underset{\sim}{A})$. This element is easily seen to depend only on the (B,f) structure of M.

Definition: If M^n is a (B,f) manifold and $U : \underset{\sim}{TB} \longrightarrow \underset{\sim}{A}$ is a Thom class, then the fundamental class of $(M,\partial M)$ is the class $[M,\partial M] \in H_n(M,\partial M;\underset{\sim}{A})$. If ∂M is empty, this class will be denoted $[M] \in H_n(M;\underset{\sim}{A})$.

If one collapses the complement of the tubular neighborhood of the boundary by

$$d : M/\partial M \longrightarrow M/\{\partial M \cup (M - \partial M \times (0,1))\} = \Sigma(\partial M/\phi)$$

one has the map which defines the boundary homomorphism in homology

$$\partial : H_s(M,\partial M;\underset{\sim}{A}) \longrightarrow H_{s-1}(\partial M;\underset{\sim}{A}).$$

If one composes the map defining $[M,\partial M]$ with the map $d \wedge 1$, it is immediate that the resulting map is the suspension of the one which defines $[\partial M]$. Thus:

Proposition: Under the boundary homomorphism in $\underset{\sim}{A}$-homology, the fundamental class of $(M,\partial M)$ is sent into the fundamental class of ∂M.

Definition: A universal characteristic class with $\underset{\sim}{A}$ coefficients for (B,f) bundles is a class $x \in H^*(B;\underset{\sim}{A})$ where $B = \lim_{r \to \infty} \{B_r, g_r\}$. If ξ is an r-plane bundle over a space X with a (B_r,f_r) structure given by a lifting $\hat{\xi} : X \longrightarrow B_r$, then the x-characteristic class of the (B_r,f_r) bundle is the class $x(\hat{\xi}) = \hat{\xi}^* \bar{g}_r^*(x) \in H^*(X;\underset{\sim}{A})$, where $\bar{g}_r : B_r \longrightarrow B$ is the usual map into the limit space. If M^n is a (B,f) manifold, the x-normal characteristic class of M is the class $x(M) \in H^*(M;\underset{\sim}{A})$ defined by $x(M) = x(\nu)$ where $\nu : M \longrightarrow B_r$ is a lift of the normal map (of some imbedding) defining the (B,f) structure on M.

Definition: If M^n is a closed (B,f) manifold and $x \in H^p(B;\underset{\sim}{A})$, then the x-characteristic number of M is the class in $H^{p-n}(pt;\underset{\sim}{A})$ obtained by evaluating $x(M)$ on the fundamental class of M. Thus if $x(M) \in H^p(M;A)$ is represented by a map $\chi : \Sigma^i(M/\phi) \longrightarrow A_{p+i}$ and $[M] \in H_n(M;\underset{\sim}{A})$ is represented by a map $\mu : S^{n+r} \longrightarrow (M/\phi) \wedge A_r$, then $x[M] = \langle x(M),[M] \rangle \in H^{p-n}(pt;\underset{\sim}{A})$ is represented by the map

$$S^{n+r+i} \xrightarrow{\Sigma^i \mu} \Sigma^i(M/\phi) \wedge A_r \xrightarrow{\chi \wedge 1} A_{p+i} \wedge A_r \xrightarrow{m_{p+i,r}} A_{p+i+r}.$$

The usefulness of characteristic numbers in cobordism theory arises from the result of L. S. Pontrjagin: Characteristic cycles on differentiable manifolds, Math. Sbor. (N.S.), 21 (63) (1947), 233-284; Amer. Math. Soc. translation 32.

Theorem: If $x \in H^p(B;\underset{\sim}{A})$ and M^n is a closed (B,f) manifold, then the x-characteristic number of M depends only on the (B,f) cobordism class of M.

Proof: Since x-characteristic numbers are clearly additive, it suffices to show that for $M = \partial W$ one has $x[M] = 0$. Letting $i : M \longrightarrow W$ be the inclusion one has $x(M) = i^*x(W)$, so

$$< x(M),[M] > = < i^*x(W),\partial[W,\partial W] > = < \delta i^*x(W),[W,\partial W] >$$

where δ is the cohomology coboundary homomorphism induced by the collapse $d : W/\partial W \longrightarrow \Sigma(\partial W/\emptyset)$. Since the cohomology sequence

$$H^p(W;\underset{\sim}{A}) \xrightarrow{i^*} H^p(\partial W;\underset{\sim}{A}) \xrightarrow{\delta} H^{p+1}(W,\partial W;\underset{\sim}{A})$$

is exact, $\delta i^*x(W) = 0$; hence $x[M] = 0$. **

Remark: Being given fibration sequences $\overline{B} \xrightarrow{h} B \xrightarrow{f} BO$, one may think of $y \in H^*(B,\overline{B};\underset{\sim}{A})$ as a relative characteristic class. Being given a (B,f) manifold M with $(\overline{B},foh)$ structure on ∂M, one has defined a relative characteristic number $y[M,\partial M] \in H^*(pt;\underset{\sim}{A})$ since the normal map gives $(M,\partial M) \longrightarrow (B,\overline{B})$. To see that such numbers are relative cobordism invariants, one may suppose by additivity that there is a (B,f) manifold W with $\partial W = M \cup (-U)$ joined along $\partial M \tilde{=} \partial U$, with U a $(\overline{B},foh)$ manifold. One

then has

$$(W,\partial W) \xrightarrow{d} \Sigma(\partial W/\emptyset) \xrightarrow{\Sigma j} \Sigma(\partial W/U) \xleftarrow{\Sigma p} \Sigma(M/\partial M)$$

giving $p_*[M,\partial M] = j_*\partial[W,\partial W]$ by the orientation assumption in the decomposition of ∂W and $y(M,\partial M) = p^*q^*y(W,U)$ where

$$(M,\partial M) \xrightarrow{p} (\partial W,U) \xrightarrow{q} (W,U)$$

so $y[M,\partial M] = < q^*y,j_*\partial[W,\partial W] > = < \delta j^*q^*y,[W,\partial W] >$. However, from the exact sequence of the triple $(W,\partial W,U)$ the composition

$$H^*(W,U) \xrightarrow{q^*} H^*(\partial W,U) \xrightarrow{\delta j^*} H^*(W,\partial W)$$

is zero, so $y[M,\partial M] = 0$. Note: Taking $\overline{B}$ empty, this reduces precisely to the closed case.

In addition to the manifold theoretic treatment of characteristic numbers by using the Thom class to construct fundamental classes, one may also give homology and cohomology theoretic descriptions of characteristic numbers which are frequently useful. In particular, these will be needed later.

As in the construction of the map $\emptyset$, one has for any r-plane bundle γ over a space X the composition

$$\gamma \xrightarrow{\Delta} \gamma \times \gamma \xrightarrow{\pi \times p} X \times T\gamma \longrightarrow (X/\emptyset)\wedge T\gamma$$

giving a map $\emptyset : T\gamma \longrightarrow (X/\emptyset)\wedge T\gamma$.

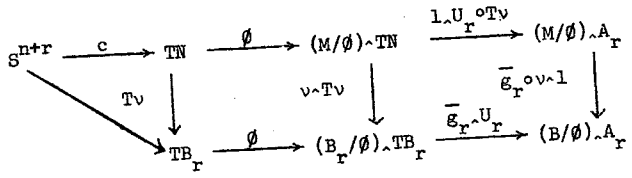
Applying this to the bundle $f_r^*(\gamma^r)$ over B_r, and composing with the Thom class and inclusion of B_r in B gives

$$(1\wedge U_r)\circ(\overline{g}_r\wedge 1)\circ\emptyset : TB_r \longrightarrow (B/\emptyset)\wedge A_r$$

inducing on homotopy the homomorphism

$$\Omega_n(B,f) = \lim_{r\to\infty} \pi_{n+r}(TB_r,^{\infty}) \longrightarrow \lim_{r\to\infty} \pi_{n+r}((B/\emptyset)\wedge A_r) = H_n(B;\underset{\sim}{A}).$$

If M^n is a closed (B,f) manifold, so that $M/\emptyset$ maps into $B_r/\emptyset$ under the normal map, one has the commutative diagram



Thus the homotopy homomorphism is the homomorphism sending the cobordism class of a manifold M^n into the image under the normal map of the fundamental class of M^n. Thus the pairing of homology and cohomology of B into the cohomology of a point gives

$$\Omega_n(B,f) \otimes H^p(B;\underset{\sim}{A}) \longrightarrow H_n(B;\underset{\sim}{A}) \otimes H^p(B;\underset{\sim}{A}) \longrightarrow H^{p-n}(pt;\underset{\sim}{A})$$

which coincides with the evaluation of characteristic numbers.

In addition, the composition

$$TB_r\wedge A_s \xrightarrow{\emptyset\wedge 1} (B_r/\emptyset)\wedge TB_r\wedge A_s \xrightarrow{1\wedge U_r\wedge 1} (B_r/\emptyset)\wedge A_r\wedge A_s \xrightarrow{1\wedge m_{r,s}} (B_r/\emptyset)\wedge A_{r+s}$$

gives rise to the Thom homomorphism in homology

$$\phi_U : \tilde{H}_j(TB_r;\underset{\sim}{A}) \longrightarrow H_{j-r}(B_r;\underset{\sim}{A})$$
$$\| \qquad\qquad\qquad\qquad \|$$
$$\lim_{s\to\infty} \pi_{j+s}(TB_r\wedge A_s) \qquad \lim_{s\to\infty} \pi_{j+s}((B_r/\emptyset)\wedge A_{r+s})$$

determined by the Thom class U. (Note: This is given by $\pi_*\circ(\cap U)$ since the map $\emptyset$ is the composition of $\pi\wedge 1$ with the map used in defining cap

products.) The homomorphism $\Omega_n(B,f) \longrightarrow H_n(B;\underset{\sim}{A})$ may then be interpreted as the composition of the Hurewicz homomorphism in $\underset{\sim}{A}$-homology $\pi_{n+r}(TB_r,^{\infty}) \longrightarrow \tilde{H}_{n+r}(TB_r;\underset{\sim}{A})$ given by $TB_r = TB_r\wedge S^0 \xrightarrow{1\wedge\alpha_0} TB_r\wedge A_0$ and the Thom homomorphism determined by U (at least after letting r go to infinity).

It is more common to consider the Thom homomorphism in cohomology theory. If one begins with the map

$$TB_r \xrightarrow{\emptyset} (B_r/\emptyset)\wedge TB_r \xrightarrow{1\wedge U_r} (B_r/\emptyset)\wedge A_r$$

and chooses a map $\chi_r : \Sigma^i(B_r/\emptyset) \longrightarrow A_{p+i}$ representing a class $x_r \in H^p(B_r;\underset{\sim}{A})$ then the composition

$$\Sigma^i(TB_r) \longrightarrow \Sigma^i(B_r/\emptyset)\wedge A_r \xrightarrow{\chi_r\wedge 1} A_{p+i}\wedge A_r \xrightarrow{m_{p+i,r}} A_{p+i+r}$$

represents a class in $\tilde{H}^{p+r}(TB_r;\underset{\sim}{A})$. This defines the cohomology Thom homomorphism

$$\Phi^U : H^p(B_r;\underset{\sim}{A}) \longrightarrow \tilde{H}^{p+r}(TB_r;\underset{\sim}{A}).$$

The construction of $\emptyset$ shows that $\Phi^U(x_r) = \pi_r^*(x_r) \cup U_r$, where $\pi_r : f_r^*(\gamma^r) \longrightarrow B_r$ is the projection and $U_r : TB_r \longrightarrow A_r$ is interpreted as a class $U_r \in \tilde{H}^r(TB_r;\underset{\sim}{A})$.

In particular, for $x \in H^p(B;\underset{\sim}{A})$ one has the sequence of elements $x_r = \bar{g}_r^*(x) \in H^p(B_r;\underset{\sim}{A})$, lifting to elements $\pi_r^*(x_r) \cup U_r \in \tilde{H}^{p+r}(TB_r;\underset{\sim}{A})$. If M^n is a closed (B,f) manifold with cobordism class represented by the map $\phi_M : S^{n+r} \longrightarrow TB_r$, then $x[M] = \phi_M^*(\pi_r^*(x_r) \cup U_r) \in \tilde{H}^{p+r}(S^{n+r};\underset{\sim}{A}) = H^{p-n}(pt;\underset{\sim}{A})$. (Note: It is immediate that this agrees with the previous homology interpretation. All interpretations are really based on the naturality of the map $\emptyset$).

Definition: The Thom class $U : \underset{\sim}{TB} \longrightarrow \underset{\sim}{A}$ is said to be an $\underset{\sim}{A}$-orientation if for each point b_r of B_r there is a framing of δ^r so that the classes determined by $U_r \circ Tb_r$ and $\alpha_r : S^r \longrightarrow A_r$ are the same in $\tilde{H}^r(S^r;\underset{\sim}{A})$.

Remarks: 1) This is the assertion that the bundles $f_r^*(\gamma^r)$ are (uniformly) $\underset{\sim}{A}$-oriented in the sense of A. Dold: Relations between ordinary and extraordinary cohomology, Notes, Aarhus Colloquium on Algebraic Topology, Aarhus, 1962.

2) If the class of $\alpha : \underset{\sim}{S} \longrightarrow \underset{\sim}{A}$ as an element of $\tilde{H}^0(S^0;\underset{\sim}{A})$ does not have order 2, this gives a preferred orientation to the fiber δ^r. If this class has order 2, then all $\underset{\sim}{A}$ cohomology is of order 2 and orientation doesn't enter the situation.

Proposition: Let $U : \underset{\sim}{TB} \longrightarrow A$ be an A-orientation, ξ an r-plane bundle over a finite complex X with $\tilde{\xi} : X \longrightarrow B_r$ a (B_r, f_r) structure on ξ. Then the Thom homomorphism

$$\Phi^U : H^p(X;\underset{\sim}{A}) \longrightarrow \tilde{H}^{p+r}(T\xi;\underset{\sim}{A})$$

is an isomorphism.

Proof: The composition $U_r \circ T\tilde{\xi} : T\xi \longrightarrow A_r$ defines an $\underset{\sim}{A}$-orientation of ξ. Over any cell D^n of X the bundle ξ is trivial, and D^n being path connected and simply connected the class $U_r \circ T\tilde{\xi}$ orients the bundle ξ over D^n. Over D^n one then has the Thom space equivalent to $D^n \wedge S^r$, with the Thom homomorphism being just the suspension isomorphism. Thus the Thom homomorphism defines a homomorphism of the $\underset{\sim}{A}$ spectral sequence of X into the reduced cohomology $\underset{\sim}{A}$ spectral sequence of $T\xi$ which is an isomorphism on E_2, hence also on E_∞. **

Corollary: If $U : \underset{\sim}{TB} \longrightarrow \underset{\sim}{A}$ is an $\underset{\sim}{A}$ orientation and M^n is a (B,f) manifold, then the Thom homomorphisms Φ^U

$$H^p(M;\underset{\sim}{A}) \longrightarrow \tilde{H}^{p+r}(TN;\underset{\sim}{A}),$$

$$H^p(\partial M;\underset{\sim}{A}) \longrightarrow \tilde{H}^{p+r}(TN';\underset{\sim}{A}), \text{ and}$$

$$H^p(M,\partial M;A) \longrightarrow \tilde{H}^{p+r}(TN,TN';\underset{\sim}{A})$$

are isomorphisms.

Let M^n be a manifold, and let M be imbedded in H^{n+r} with $\partial M \subset R^{n+r-1}$ in the usual way, and let ν be the normal bundle of M, ν' the normal bundle of ∂M.

Theorem: (Atiyah [14]) The pairs

a) $(M/\partial M)$ and $T\nu$, or

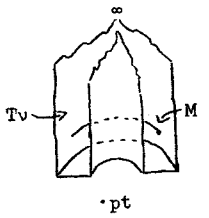
b) $(M/\emptyset)$ and $(T\nu/T\nu')$

are dual in S^{n+r+1}.

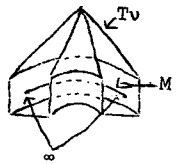
[B and C in S^k are dual if B and C are disjoint and each is a strong deformation retract of the complement of the other.]

Proof: Let N be a tubular neighborhood of M in H^{n+r}. Consider S^{n+r+1} as $H^{n+r} \times R$ with $(R^{n+r-1} \cup \infty) \times R \cup H^{n+r} \times \{-\infty,\infty\}$ collapsed to a base point ∞.

Consider $(T\nu/T\nu')$ as $(N \times 0 \cup \partial N \times [0,\infty) \cup \infty)$ and then one may collapse the complement which is $\{H^{n+r} \times (-\infty,0) \cup (-N) \times [0,\infty)\} \cup \text{interior}(N) \times (0,\infty)$ to pt $\cup$ $(M-\partial M) \times 1$ by a strong deformation retract, and conversely, the complement of a point inside $H^{n+r} \times (-\infty,0) \cup (-N) \times [0,\infty)$ and $(M-\partial M) \times 1$ may be collapsed onto $(T\nu/T\nu')$ by a strong deformation retraction.



Considering $(M/\partial M)$ as $(M \times 1) \cup \infty$ one may collapse the complement by a strong deformation retract onto the subset $\bar{N} \times 0 \cup \partial\bar{N} \times [0,2] \cup c$, where c is the cone on $\partial\bar{N} \times 2$ with vertex some point with last coordinate larger than 2, where $\bar{N} = N - N \cap R^{n+r-1}$. This subset is clearly just $T\nu$. Similarly, removal of this set $T\nu$ gives a set collapsing onto $M/\partial M$.



Note: All deformations are obtained by radial deformations toward points in question, and scalar multiplication expansions in the fibers of the normal bundles of M. **

If $B,C \subset S^k$ are disjoint sets as above, let $p_0 \in S^k - B \cup C$ and apply stereographic projection to map B, C into R^k as disjoint subsets. Let $f : B \times C \longrightarrow S^{k-1}$ by $f(b,c) = (b-c)/\|b-c\|$. Letting b, c be base points of B, C respectively, $f(B \times c) \cup f(b \times C)$ is a proper subset of S^{k-1} and f factors homotopically through a map $\bar{f} : B \wedge C \longrightarrow S^{k-1}$.

One then has defined a duality as follows: For $\alpha \in \tilde{H}_p(B;\underset{\sim}{A})$ choose a representative map $a : S^{p+i} \longrightarrow B \wedge A_i$ and then

$$S^{p+i} \wedge C \xrightarrow{a \wedge 1} B \wedge A_i \wedge C \xrightarrow{\bar{f} \wedge 1} S^{k-1} \wedge A_i \longrightarrow A_{i+k-1}$$

defines a class in $\tilde{H}^{k-1-p}(C;\underset{\sim}{A})$, denoted $D\alpha$.

Lemma: Let $b \in B \subset R^k$ with B an imbedded disc. Then $H^*(R^k-b, R^k-B;\underset{\sim}{A}) = 0$.

Proof: $H^*(R^k-b, R^k-B) \cong H^*(S^k-b, S^k-B)$ by excision. Since S^k-b and S^k-B are contractible, $\tilde{H}^*(S^k-b) = \tilde{H}^*(S^k-B) = 0$ and the exact sequence of the pair gives the result. **

Corollary: If $B \subset R^k$ is an imbedded disc then $D : H_p(R^k, R^k-B;\underset{\sim}{A}) \longrightarrow H^{k-p}(B;\underset{\sim}{A})$ is an isomorphism.

Proof: One has the diagram

$$\begin{array}{ccccccc}
H_p(R^k-b, R^k-B) & \longrightarrow & H_p(R^k, R^k-B) & \longrightarrow & H_p(R^k, R^k-b) & \longrightarrow & H_{p-1}(R^k-b, R^k- \\
\downarrow{\scriptstyle D'} & & \downarrow{\scriptstyle D} & & \downarrow{\scriptstyle D''} & & \downarrow{\scriptstyle D'} \\
H^{k-p}(B,b) & \longrightarrow & H^{k-p}(B) & \longrightarrow & H^{k-p}(b) & \longrightarrow & H^{k-p+1}(B,b)
\end{array}$$

in which the end maps are both isomorphisms since the groups are zero, and complete the proof one need only check D'', which is clearly an isomorphi

Theorem: (Alexander Duality) For any polyhedral pair (B,B') in R^k, $D : H_p(R^k-B', R^k-B;\underset{\sim}{A}) \longrightarrow H^{k-p}(B,B';\underset{\sim}{A})$ is an isomorphism.

Proof: By naturality it suffices to consider the case $B' = \emptyset$, and t one may apply a Mayer-Vietoris argument using induction on the number of cells of B, the corollary and the five lemma. **

Theorem: (Spanier-Whitehead duality) For any pair $B,C \subset S^k$ as above $D : \tilde{H}_p(B;\underset{\sim}{A}) \longrightarrow \tilde{H}^{k-1-p}(C;\underset{\sim}{A})$ is an isomorphism.

Proof: D is given by the composition of isomorphisms

$$\tilde{H}_p(B;\underset{\sim}{A}) \cong \tilde{H}_p(S^k-C;\underset{\sim}{A}) \xleftarrow{\cong} H_{p+1}(S^k-c, S^k-C;\underset{\sim}{A}) \xleftarrow{\cong} H_{p+1}(R^k-c, R^k-C;\underset{\sim}{A}) \xrightarrow{D} \tilde{H}^{k-p-1}(C;\underset{\sim}{A}).$$

Note: See Spanier [110], pages 295 and 462.

Theorem: (Hsiang and Wall [58]) A class $\alpha \in \tilde{H}^r(T\nu;\underset{\sim}{A})$ is a Dold orientation if and only if the class $D^{-1}(\alpha) \in H_n(M,\partial M;\underset{\sim}{A})$ is an $\underset{\sim}{A}$ orientat in the sense of Whitehead; i.e. for each point $q \in M - \partial M$ the class in $H^n(M,\partial M;\underset{\sim}{S})$ obtained by collapsing $M/\partial M$ onto $D^n/\partial D^n$ where D^n is a disc

neighborhood of q and the class $D^{-1}(\alpha)$ have Kronecker product the class defined by the unit in $H^0(pt;\underset{\sim}{A})$.

Proof: The map $D^n \longrightarrow M - \partial M$ given by taking the disc neighborhood of q defines the collapse $M/\partial M \longrightarrow D^n/\partial D^n$ which is patently dual to the inclusion of TD^n in $T\nu$, but TD^n is homotopy equivalent to the Thom space of a fiber. **

Note: Similarly, the collapse $S^{n+r} \longrightarrow T\nu/T\nu'$ is dual to the map of M to a point.

Corollary: If the Thom class $U : \underset{\sim}{TB} \longrightarrow \underset{\sim}{A}$ is an $\underset{\sim}{A}$ orientation, then for any (B,f) manifold M^n, the fundamental class $[M,\partial M] \in H_n(M,\partial M;\underset{\sim}{A})$ is an $\underset{\sim}{A}$ orientation in the sense of Whitehead.

Theorem: (Poincaré-Lefschetz duality) If $U : \underset{\sim}{TB} \longrightarrow \underset{\sim}{A}$ is an $\underset{\sim}{A}$-orientation, then for any (B,f) manifold M^n one has isomorphisms

$$H^q(M;\underset{\sim}{A}) \longrightarrow H_{n-q}(M,\partial M;\underset{\sim}{A})$$

$$H^q(M,\partial M;\underset{\sim}{A}) \longrightarrow H_{n-q}(M;\underset{\sim}{A})$$

given by the cap-product with the fundamental class $[M,\partial M]$.

The cap-product relation is given by

$$u:S^{n+r} \xrightarrow{c} TN/TN' \xrightarrow{\emptyset} (M/\partial M)\wedge TN \xrightarrow{1\wedge U_r\circ T\nu} (M/\partial M)\wedge A_r \xrightarrow{\Delta\wedge 1} (M/\partial M)\wedge(M/\emptyset)\wedge A_r$$

where Δ is the map given from the diagonal. If $\Sigma^i(M/\emptyset) \xrightarrow{x} A_{q+i}$ represents a class in $H^q(M;\underset{\sim}{A})$, the cap product is represented by

$$S^{n+r+i} \xrightarrow{\Sigma^i u} (M/\partial M)\wedge\Sigma^i(M/\emptyset)\wedge A_r \xrightarrow{1\wedge x\wedge 1} (M/\partial M)\wedge A_{q+i}\wedge A_r \xrightarrow{1\wedge m_{q+i,r}} (M/\partial M)\wedge A_{r+q+i}.$$

A similar formula defines the other homomorphism.

Proof: These isomorphisms are just the composite of Spanier-Whitehead duality and the Thom isomorphisms. **

Chapter IV

The Interesting Examples - A Survey of the Literature

Since cobordism theory is a classificational tool, the interest really lies in the investigation of specific classification problems. Numerous examples have been considered and hence a vast literature exists, with few really central theoretical tools, largely due to the idiosyncrasies inherent in the examples. The purpose of this chapter is to list many of these examples and indicate briefly what is known and where to find it in the literature.

Example 1: Framed cobordism: Ω_*^{fr}.

Historically: First application of cobordism theory, intended to study the homotopy of spheres.

Objects: Framed manifolds, i.e. manifolds with an equivalence class of trivializations of the normal bundle.

Determination: (B,f) cobordism with each B_r contractible (classifyi space for the identity subgroup 1_r of 0_r), so $\Omega_n^{fr} \tilde{=} \lim_{r\to\infty} \pi_{n+r}(S^r)$ is the stable homotopy of spheres (Pontrjagin [101]).

Results: A vast literature exists but is largely unrelated to cobordis Use of surgery (Milnor [83], Wallace [137]) to construct framed cobordisms shows that representatives frequently may be taken to be homotopy spheres (Kervaire-Milnor [61]). Recent work of Conner and Floyd [41] has placed the e-invariant of Adams [4] in a cobordism framework.

Example 2: Unoriented cobordism: $\mathcal{N}_*$.

Historically: The turning point for cobordism theory.

Objects: All compact manifolds, i.e. the category $(\mathcal{O},\partial,i)$.

Determination: Equivalent to (B,f) cobordism with $B_r = BO_r$ and f_r the identity map. (Thom [127]).

Calculation: π_* is the polynomial ring over Z_2 on classes x_i of dimension i for each integer i not of the form 2^s-1. Even dimensional generators may be taken to be the classes of real projective spaces. (Thom [127]). Odd dimensional generators were constructed by Dold [43].

Characteristic numbers: Z_2 cohomology characteristic numbers give complete invariants (Thom [127]). All relations among these numbers (expressed tangentially) are given by Wu's formulae (Wu [142]) relating to the action of the Steenrod algebra (Dold [44]).

Example 3: Complex cobordism: Ω^U_*.

Objects: Stably almost complex manifolds - manifolds with an equivalence class of complex vector bundle structures on the normal bundle.

Determination: (B,f) cobordism with $B_{2r} = B_{2r+1} = BU_r$ the classifying space for the unitary group U_r (limit of complex Grassmann manifolds).

Calculation: Ω^U_* is the integral polynomial ring on classes x_i of dimension $2i$ for each integer i, with x_i represented by a projective complex algebraic variety (Milnor [82], Novikov [92, 93]). In fact, every class is represented by such a variety (Milnor; see Hirzebruch [54] or Thom [129]).

Characteristic numbers: Cobordism is determined by integral cohomology characteristic numbers (Milnor [82]). All relations among these numbers are given by the Atiyah-Hirzebruch [17] form of Riemann-Roch theorem relating complex K-theory to rational cohomology (Stong [117], Hattori [52]).

Relation to π_*: Ω^U_* maps onto the squares of classes of π_* (Milnor [87]).

Relation to Ω^{fr}_*: Every framed manifold bounds a complex manifold. The Todd class homomorphism $\Omega_*((BU,f),(Bl,f)) \longrightarrow Q$ of the relative cobordism group induces the Adams e homomorphism (Conner and Floyd [41]).

Example 4: Oriented cobordism: Ω^{SO}_*.

Objects: Oriented manifolds.

Determination: (B,f) cobordism with $B_r = BSO_r$ the classifying space for the special orthogonal group SO_r (limit of Grassmannians of oriented planes) (Thom [127]).

Calculation: $\Omega^{SO}_* \otimes Q$ is the rational polynomial ring on classes x_{4i} of the complex projective spaces $CP(2i)$ (Thom [127]). Ω^{SO}_* has no odd torsion (Milnor [81], Averbuh [21]) and Ω^{SO}_*/Torsion is a polynomial ring over Z on $4i$ dimensional generators (Milnor [81], Novikov [92,93]). Ω^{SO}_* has only torsion of order two and the quotient $\Omega^{SO}_*/2\Omega^{SO}_*$ may be described as follows: Let W_* be a Z_2 polynomial ring on classes x_{2k}, x_{2k-1}, for k not a power of 2, and $x_{2^j}{}^2$. Let $\partial_1 : W_* \longrightarrow W_*$ be a derivation given by $\partial_1 x_{2k} = x_{2k-1}$, $\partial_1 x_{2k-1} = 0$, $\partial_1 x_{2^j}{}^2 = 0$. Then $\Omega^{SO}_*/2\Omega^{SO}_* \tilde{=}$ kernel ∂_1 and the image of the torsion will be the image of ∂_1 (Wall [130]).

Characteristic numbers: Cobordism is determined by Z and Z_2 cohomology, all relations among the Z_2 numbers being given by the relations of Wu together with the vanishing of the first Stiefel-Whitney class (Wall [130]). All relations among the Z numbers are given by the Riemann-Roch theorem (Stong [117]).

Relation to Ω^U_*: Ω^U_* maps onto Ω^{SO}_*/Torsion (Milnor [82]).

Relation to π_*: $\Omega^{SO}_*/2\Omega^{SO}_*$ is mapped isomorphically to the subring kernel ∂_1 described above, the x_i being (well-chosen) generators of π_* (Wall [130]).

Example 5: w_1 spherical cobordism: $\mathcal{W}_*$

Historically: This cobordism theory arises in Wall's determination of oriented cobordism, and was completely determined by Wall [130].

Objects: Manifolds for which the first Stiefel-Whitney class w_1 is the reduction of an integral cohomology class; is induced by a map into the sphere S^1.

Determination: (B,f) cobordism with B_r the total space of the fibration over $BO_r \times S^1$ induced from the path fibration over $K(Z_2,1)$ by the map realizing the cohomology class $w_1 \otimes 1 + 1 \otimes \sigma$, $\sigma \in H^1(S^1;Z_2)$ being the generator.

Calculation: Given by the polynomial ring $\mathcal{W}_*$ described above.

Characteristic numbers: Z_2 cohomology determines cobordism, all relations being given by those of Wu together with the vanishing of w_1^2.

Relation of $\mathcal{N}_*$ and Ω_*^{SO}: Maps monomorphically into $\mathcal{N}_*$, with ∂_1 describing the image of Ω_*^{SO} as above.

Example 6: Bordism: $\Omega_*(B,f)[X,A]$.

Objects: Let $F : (B,f) \longrightarrow \mathfrak{X}$ be the forgetful functor from the cobordism category of (B,f) manifolds to the category of topological spaces which takes the underlying topological space. One then has the cobordism category of (B,f) manifolds 'over' a space X. If $A \subset X$ is a subspace one has a functor $J : (B,f)/A \longrightarrow (B,f)/X$ induced by the inclusion.

Determination: $(B,f)/X$ cobordism is just the cobordism theory based on the fibration $B_r \times X \xrightarrow{\pi} B_r \xrightarrow{f_r} BO_r$, π-being the projection. The relative bordism group of the pair (X,A) is $\Omega_n(B,f)[X,A] = \Omega_n(J,\alpha)$, where α is the piecing together previously described, and is given by $\lim_{r\to\infty} \pi_{n+r}(TB_r \wedge (X/A)) = H_n(X,A;\underset{\sim}{TB})$.

Historically: These groups were originally defined by Atiyah [13], who called them the (B,f) bordism groups of the pair (X,A). He reserved the name cobordism for the dual cohomology theory with coefficients in the spectrum $\underset{\sim}{TB}$.

$\mathcal{N}_*(X,A)$: The unoriented bordism of a pair (X,A) is essentially trivial, being isomorphic as $\mathcal{N}_*$ module to $\mathcal{N}_* \otimes H_*(X,A;Z_2)$. Cobordism is determined by Z_2 cohomology. (See Conner and Floyd [36]). Operations in this theory were determined by Landweber [63, 64]. Used by Brown and Peterson [29] to study relations among Stiefel-Whitney classes.

$\Omega_*^{SO}(X,A)$: Studied extensively by Conner and Floyd [36]. One has $\Omega_*^{SO}(X,A)$ isomorphic to $H_*(X,A;\Omega_*^{SO})$ modulo the Serre class of finite groups of odd order. If all torsion of $H_*(X;Z)$ is of order 2, then Z and Z_2 cohomology characteristic numbers determine cobordism class in $\Omega_*^{SO}(X)$. Künneth theorems for this homology theory are studied in Landweber [65]. An interesting application is the use of $\Omega_*^{SO}(BU)$ in the proof of the Atiyah-Singer index theorem (Atiyah-Singer [20], Palais [97]).

$\Omega_*^{U}(X,A)$: Studied by Conner and Floyd [35, 37]. If X has no torsion in its homology $\Omega_*^{U}(X) \cong \Omega_*^{U} \otimes H_*(X;Z)$ and cobordism is determined by integral cohomology characteristic numbers. The relation of $\Omega_*^{U}(X)$ to the complex K-theory of X is studied in Conner and Floyd [41]. Operations in Ω_*^{U} theory are studied in Novikov [96].

Example 7: Special unitary cobordism: Ω_*^{SU}.

Objects: Manifolds with an equivalence class of special unitary structures on the normal bundle.

Determination: (B,f) cobordism in which $B_{2r} = B_{2r+1} = BSU_r$ is the classifying space for the special unitary group SU_r.

Calculation: First partial results were by Novikov [93]. The main structure was determined by Conner and Floyd [39], who proved that Ω_n^{SU} is torsion free except for $n \equiv 1$ or 2 modulo 8, and $\Omega_{8k+1}^{SU} \cong \Omega_{8k+2}^{SU}$ is a Z_2 vector space whose dimension is the number of partitions of k. The multiplicative structure has been described by Wall [135].

Characteristic numbers: KO-theory characteristic numbers determine cobordism class (Anderson, Brown, and Peterson [6]). Ignoring torsion, integral cohomology suffices, and all relations among these follow from appropriate Riemann-Roch theorems (Stong [117]).

Relation to Ω_*^U: The kernel of $\Omega_*^{SU} \longrightarrow \Omega_*^U$ is the torsion subgroup. The image was described by Conner and Floyd.

Relation to Ω_*^{fr}: The image of Ω_*^{fr} in Ω_*^{SU} is zero except in dimensions $8k+1$ and $8k+2$ where it is Z_2 (Anderson, Brown, and Peterson [6]).

Relation to $\mathcal{H}_*$: The image of Ω_*^{SU} in $\mathcal{H}_*$ is the squares of classes containing an oriented manifold all of whose Pontrjagin numbers divisible by $\mathcal{P}_1$ are even. (Conner and Landweber [42]).

Example 8: c_1 spherical manifolds: $\mathcal{W}_*^U$.

Historically: The analog in the complex case of $\mathcal{W}_*$, used by Conner and Floyd in computation of Ω_*^{SU}.

Objects: Stably almost complex manifolds for which the first Chern class c_1 is induced by a map into a sphere.

Determination: (B,f) cobordism in which $B_{2r} = B_{2r+1}$ is the total space of the fibration over $S^2 \times BU_r$ induced from the path fibration over $K(Z,2)$ by the map realizing $\sigma \otimes 1 - 1 \otimes c_1$.

Calculation: Conner and Floyd determine $\mathcal{W}_*^U$ in [39].

Characteristic numbers: Integral cohomology characteristic numbers determine cobordism class.

Relation to Ω_*^U: $\mathcal{W}_*^U$ maps monomorphically into Ω_*^U with image all classes for which numbers divisible by c_1^2 are zero.

Relation to $\mathcal{H}_*$: $\mathcal{W}_*^U$ has image in $\mathcal{H}_*$ precisely the squares of classes of $\mathcal{W}_*$ (Stong [118], Conner and Landweber [42]).

Example 9: Spin cobordism: Ω_*^{Spin}.

Objects: Manifolds with an equivalence class of Spin structures on the normal bundle. ($Spin_n$ is the simply connected covering group of SO_n; see Atiyah, Bott, and Shapiro [16] and Milnor [85]).

Determination: (B,f) cobordism with B_r the classifying space of $Spin_r$; i.e. the two-connective covering space of BSO_r.

Calculation: Preliminary results were by Novikov [93]. The main calculation is due to Anderson, Brown, and Peterson [7, 8] who showed that all torsion is of order 2, being of two types: that arising by products with a framed S^1 (similar to the SU case) and that which maps monomorphically into unoriented cobordism. Ω_*^{Spin}/Torsion is the subring of an integral polynomial ring on classes x_{4i} (dimension $4i$) consisting of all classes of dimension a multiple of 8 and twice the classes whose dimension is not a multiple of 8.

Characteristic numbers: Cobordism is determined by Z_2 cohomology and KO-theory characteristic numbers. The relations in integral cohomology all follow from the Riemann-Roch theorem.

Relation to Ω_*^{fr}: The image of framed cobordism is the same as the image of framed in SU cobordism.

Relation to Ω_*^{SO}: The kernel of the map $\Omega_*^{Spin} \longrightarrow \Omega_*^{SO}$ is in dimensions $8k + 1$ and $8k + 2$ only and is the part generated by framed manifolds.

Relation to $\mathcal{N}_*$: The image in $\mathcal{N}_*$ is all classes for which the Z_2 characteristic numbers divisible by w_1 and w_2 are zero. Preliminary work in this direction was done by Milnor [87], P. G. Anderson [9], and Stong [118], showing that the class of the square of an oriented manifold is the class of a Spin manifold.

Example 10: Spin^c, Pin, and Pin^c cobordism: $\Omega_*^{Spin^c}$, Ω_*^{Pin}, and $\Omega_*^{Pin^c}$.

Objects: Manifolds with an equivalence class of Spin^c, Pin, or Pin^c structures (see Atiyah, Bott, and Shapiro [16]).

Determination: (B,f) cobordism with the obvious classifying spaces. (BSpin^c is BSO with w_2 made reduced integral, BPin^c is BO with w_2 made reduced integral, and BPin is BO with w_2 killed).

Calculation: Due to Anderson, Brown, and Peterson (announced in [8]), largely as a byproduct in the study of Spin. $\Omega_*^{Spin^c}$/Torsion is discussed in Stong [117].

Remarks: Results have not yet been published for Pin aand Pin^c. These groups are 2 primary, having elements of arbitrarily large order. Images in $\mathcal{N}_*$ are those classes for which the appropriate Stiefel-Whitney numbers vanish.

Note: Pin gives the first example of a theory for which the tangential and normal structures are not of the same type. Specifically, if M has a Pin normal bundle, then the tangent bundle has $w_2 = w_1^2$ and hence the tangent bundle does not necessarily have a Pin structure.

Example 11: Complex-Spin cobordism: Ω_*^{c-s}.

Objects: Manifolds with both a stably almost complex and a Spin structure.

Determination: (B,f) cobordism with B the fibration over BU induced from the fibration of BSpin over BSO.

Calculation: Ω_n^{c-s} is the direct sum of Ω_n^{SU} and a free abelian group (Stong [121]).

Remark: Useful in trying to understand the relationship between SU and Spin cobordism.

Example 12: Symplectic cobordism: Ω_*^{Sp}.

Objects: manifolds with an equivalence class of quaternionic vector bundle structures on the normal bundle.

Determination: (B,f) cobordism with $B_{4r} = B_{4r+1} = \ldots = B_{4r+3}$ the classifying space BSp_r of the symplectic group Sp_r of unitary quaternionic $r \times r$ matrices (limit of quaternionic Grassmann manifolds).

Calculation: Novikov [93] showed that $\Omega_*^{Sp} \otimes Z[1/2]$ is polynomial on $4i$ dimensional generators and calculated the low dimensional groups. Liulevicius [75] calculated more low groups using the Adams spectral sequence, and computations are still in progress using that method.

Relation with $\mathcal{N}_*$: The image in unoriented cobordism is zero in dimensions less than 24 (Stong [123]).

Remark: The corresponding bordism theory is studied in Landweber [68].

Example 13: Quasi-symplectic cobordism.

Objects: Manifolds for which the normal bundle is a sum of tensor products of quaternionic vector bundles. (Note: The tensor product of quaternionic bundles in only a real bundle.)